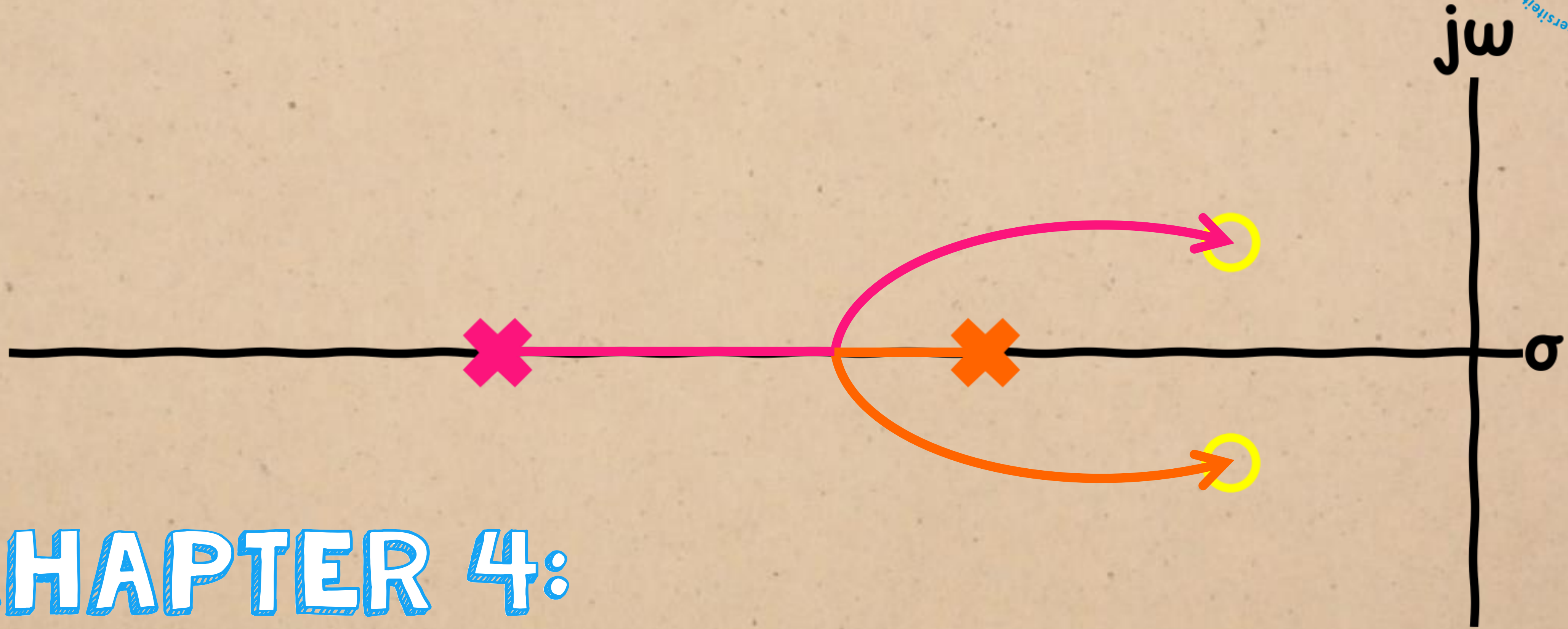
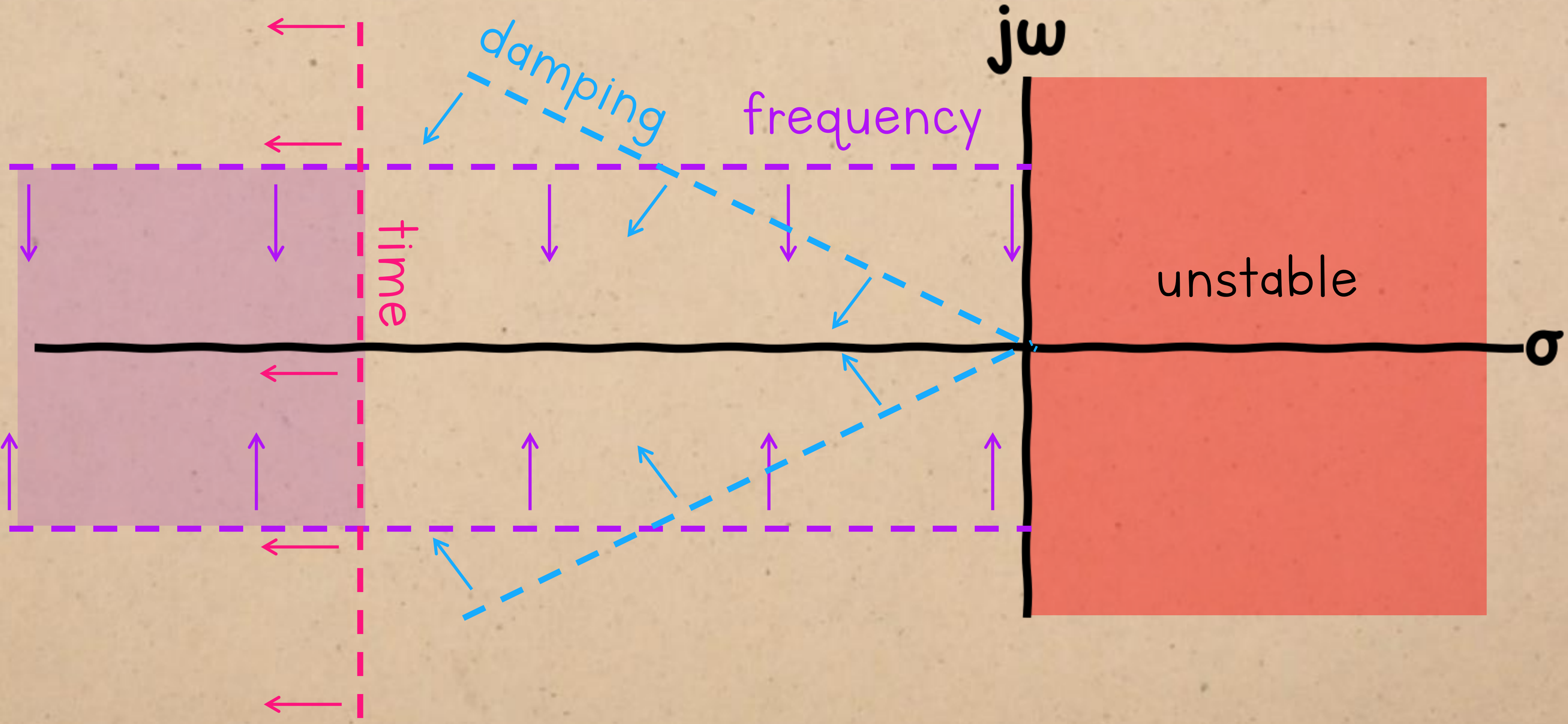
UCT MECHATRONICS



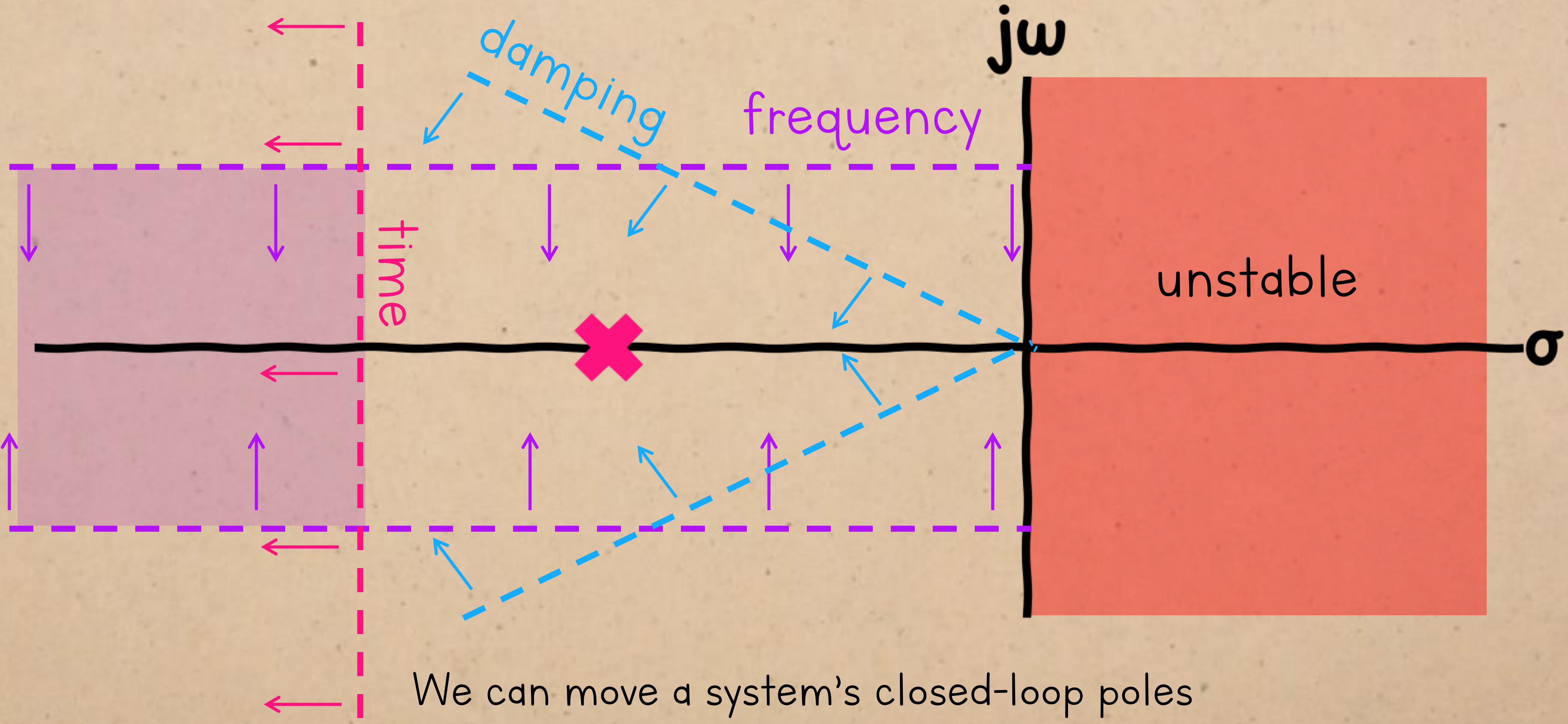
CHAPTER 4:

ROOT LOCUS I

LAST TIME:

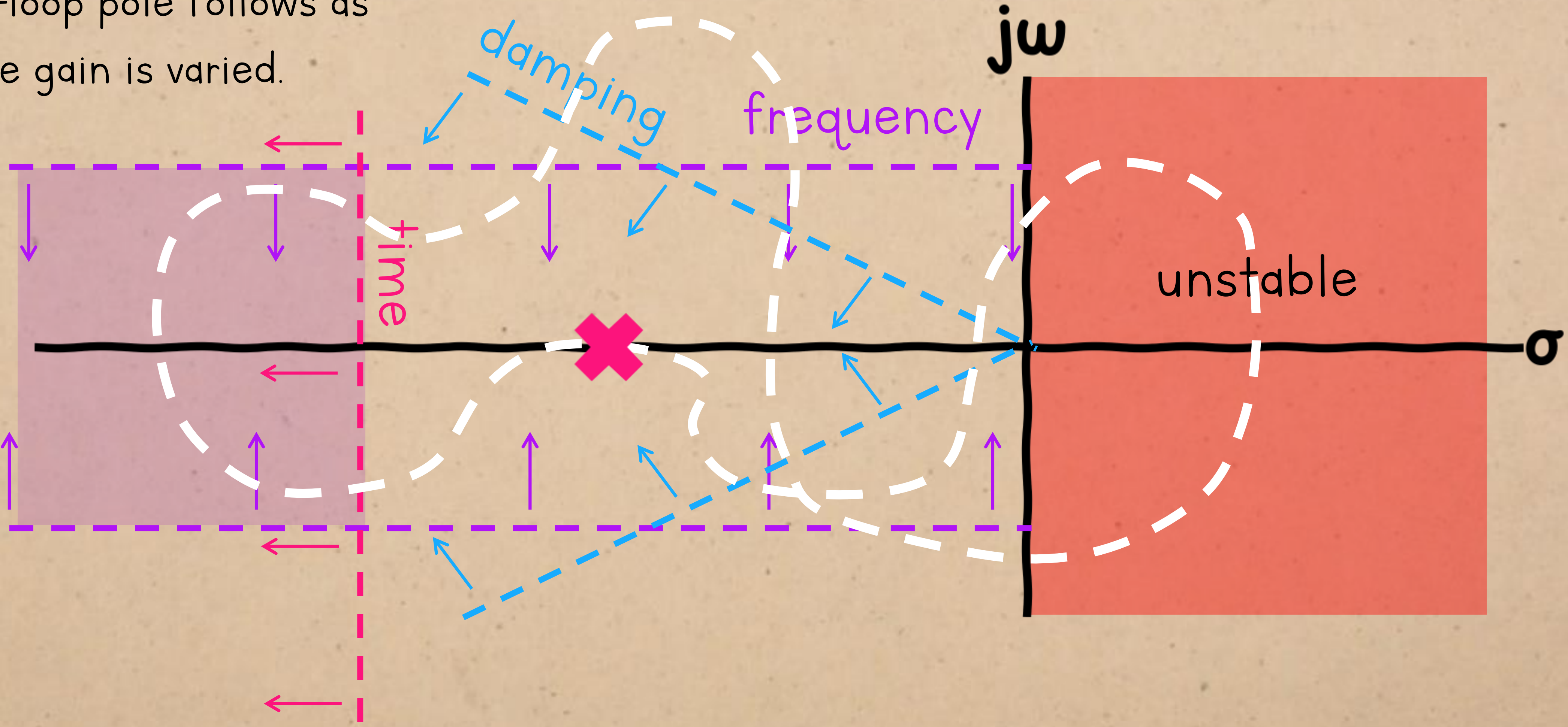


LAST TIME:



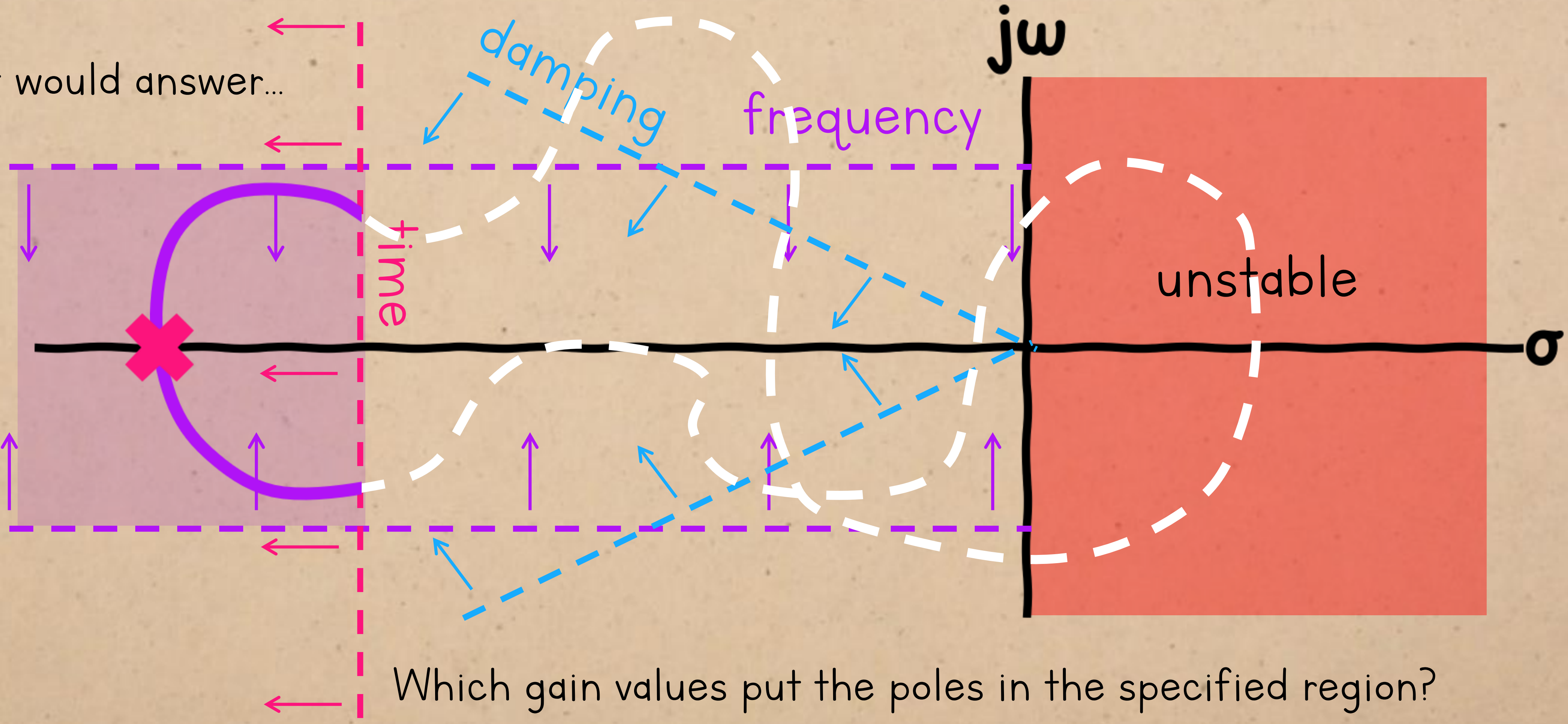
We can move a system's closed-loop poles
by changing the open-loop gain.

It would be useful to know the path that each closed-loop pole follows as the gain is varied.



*it won't look like this one.

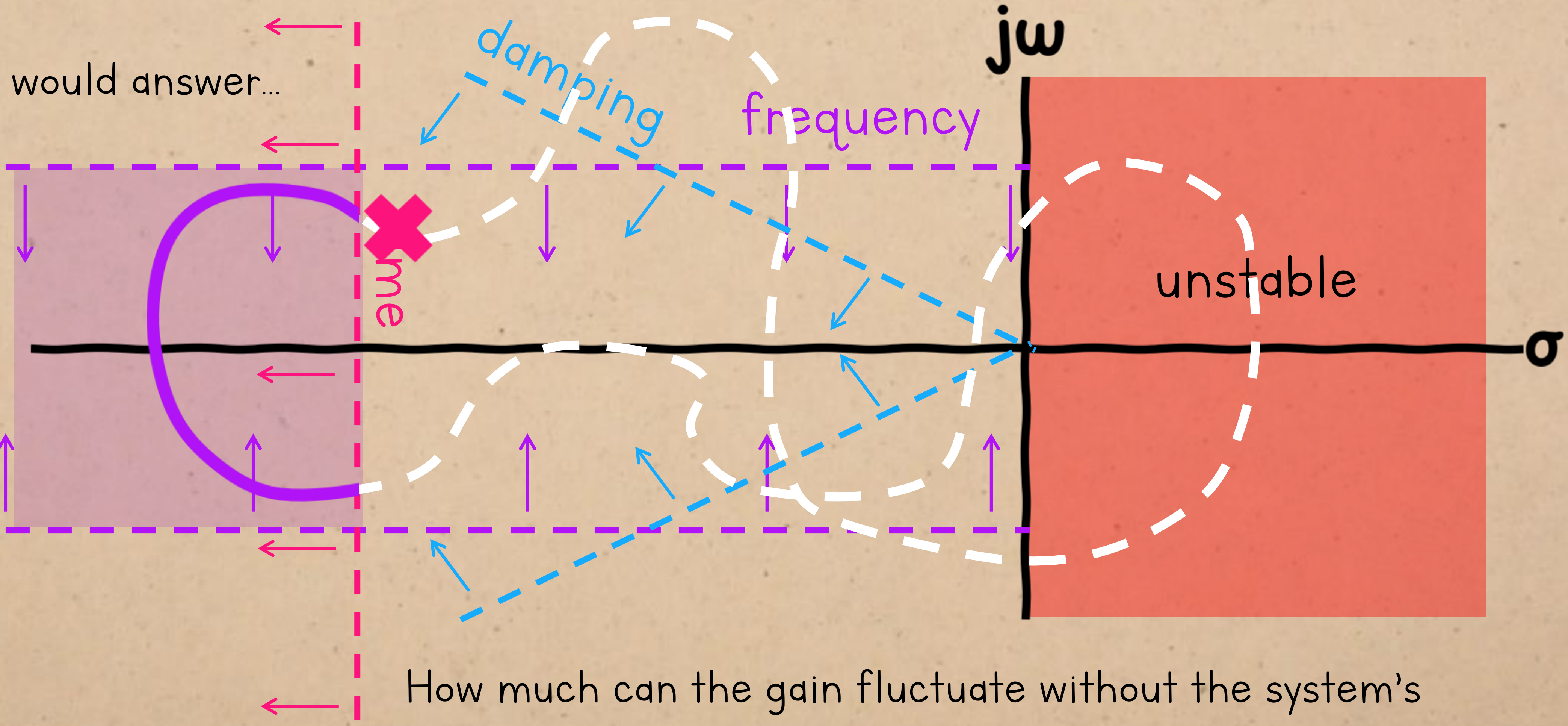
That would answer...



Which gain values put the poles in the specified region?

*it won't look like this one.

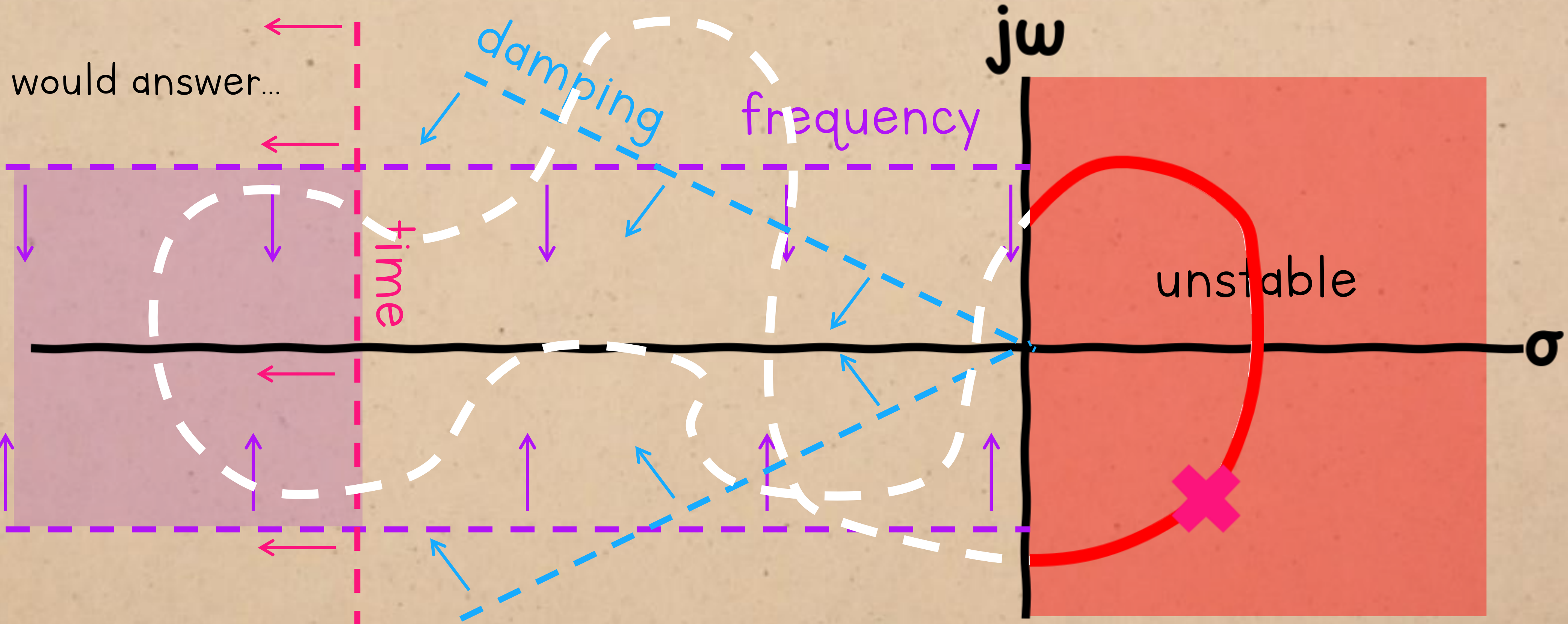
That would answer...



How much can the gain fluctuate without the system's behaviour failing to meet specifications?

*it won't look like this one.

That would answer...

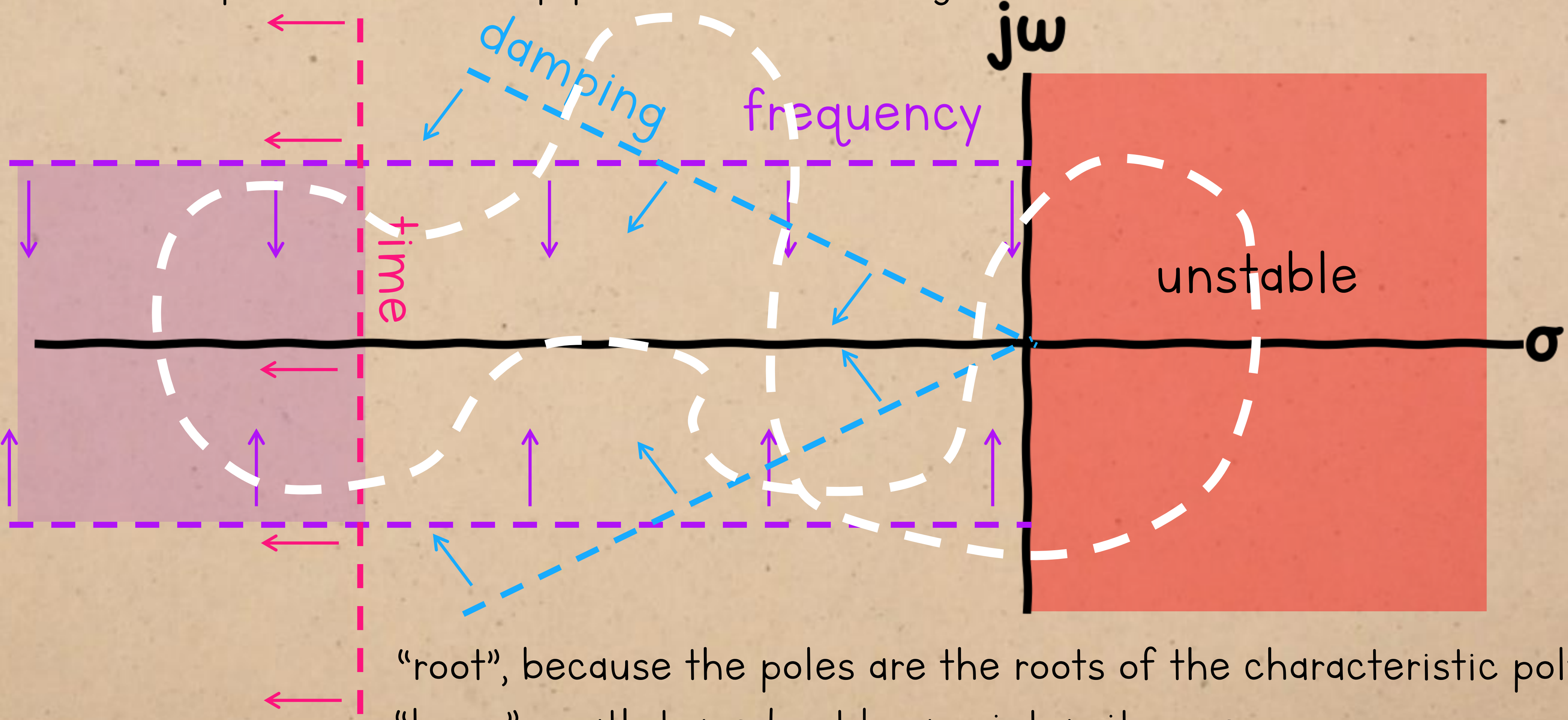


Are there any gains that would destabilize the system?

*it won't look like this one.

ROOT LOCUS

= the path a closed-loop pole follows as the gain is increased from 0 to ∞

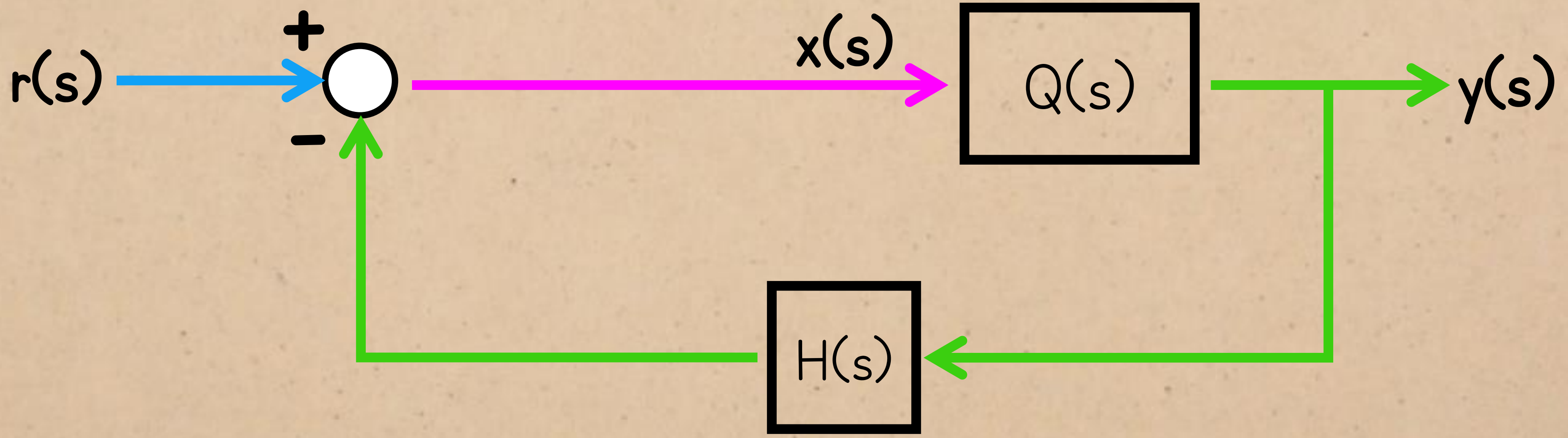


"root", because the poles are the roots of the characteristic polynomial

"locus" = path traced out by a point as it moves

ROOT LOCUS

How does the open-loop gain affect the closed-loop transfer function?



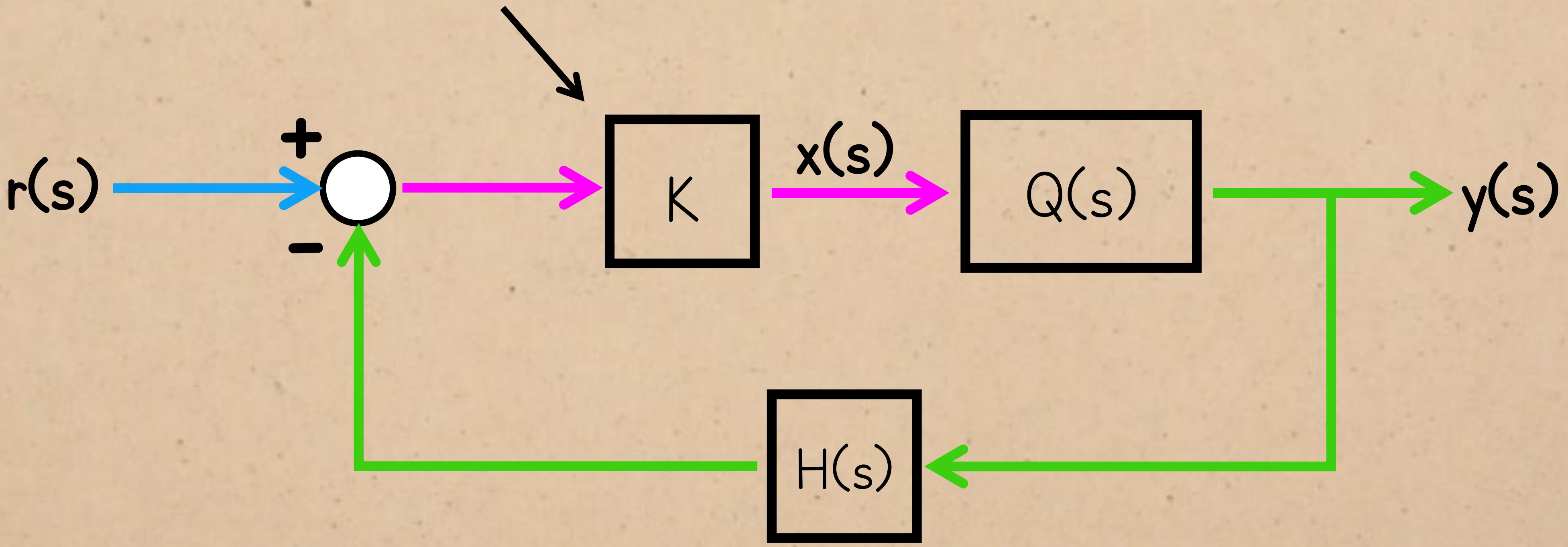
"**Gain**" can refer to any scalar

e.g. the gain of a transfer function

$$\lim_{s \to 0} Q(s)$$

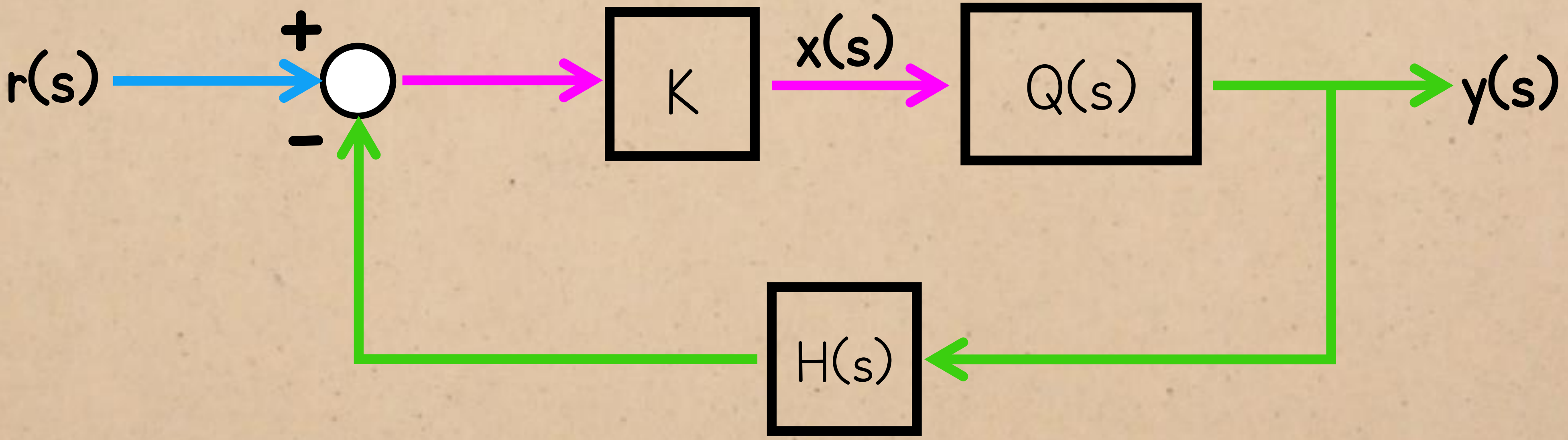
ROOT LOCUS

...But ***the* gain** of the system is the gain of the forward path:



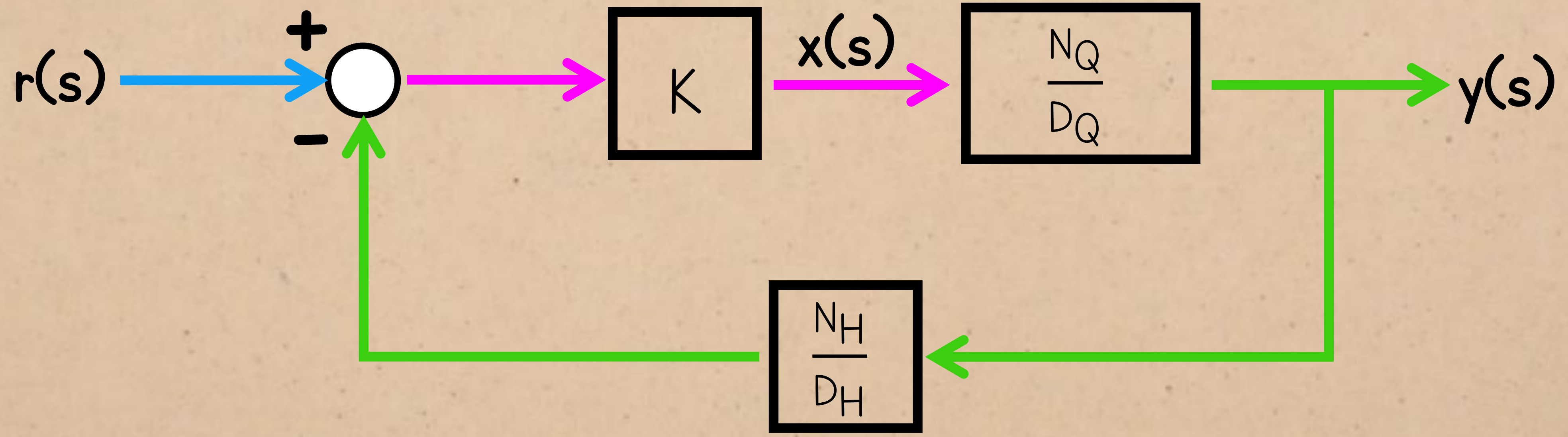
ROOT LOCUS

$$G_{CL} = \frac{Q(s)}{1 + Q(s)}$$



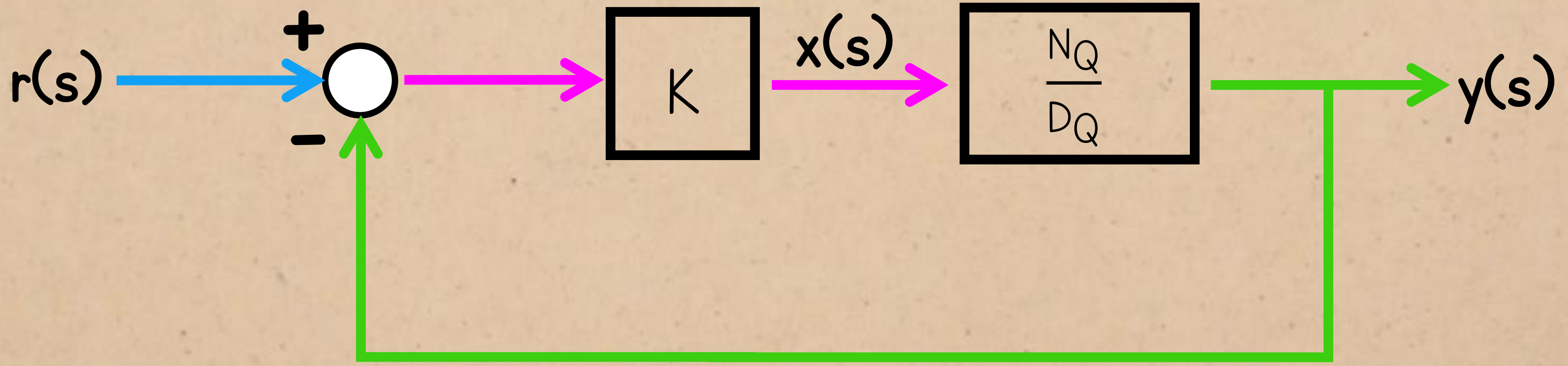
ROOT LOCUS

$$G_{CL} = \frac{KN_QD_H}{KN_QN_H + D_QD_H}$$



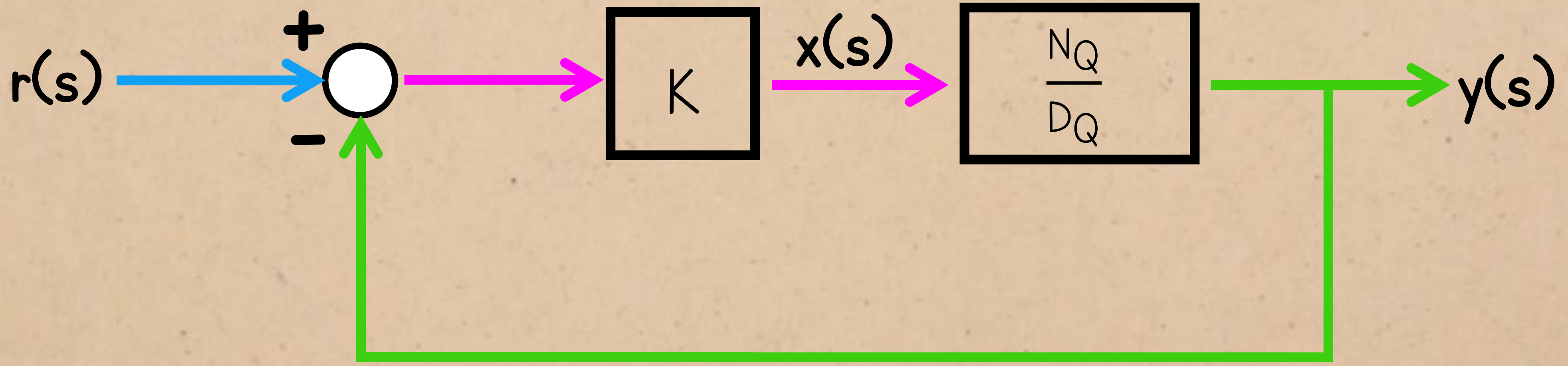
ROOT LOCUS

$$G_{CL} = \frac{KN_Q}{KN_Q + D_Q}$$ (assuming unity feedback)



ROOT LOCUS

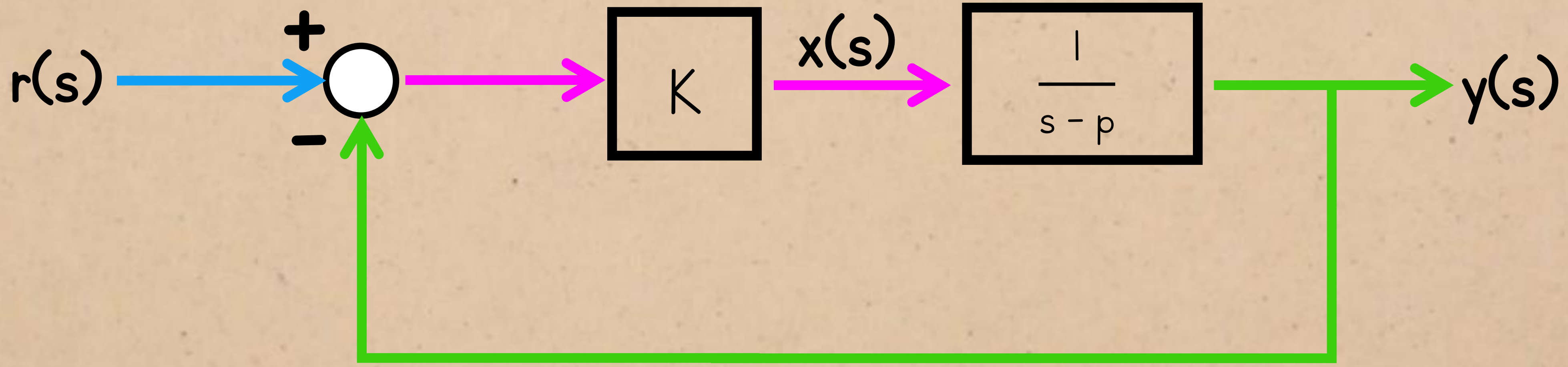
characteristic equation: $KN_Q + D_Q = 0$



ROOT LOCUS

First-order system:

characteristic equation: $K + s - p = 0$



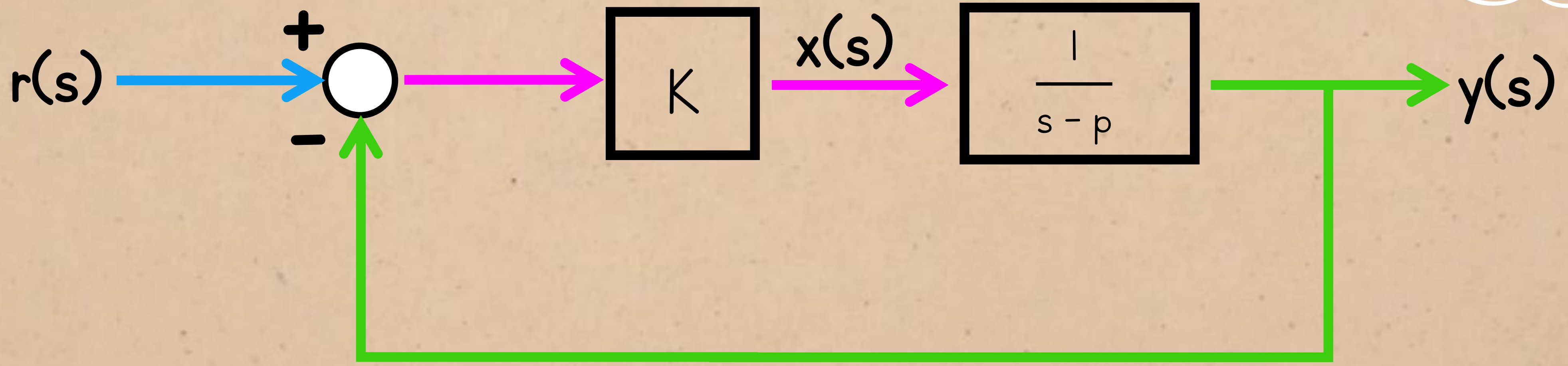
ROOT LOCUS

I'm not using this form for the general first-order system when we look at the root locus, because...

$$\frac{A}{\tau s - p}$$

First-order system:

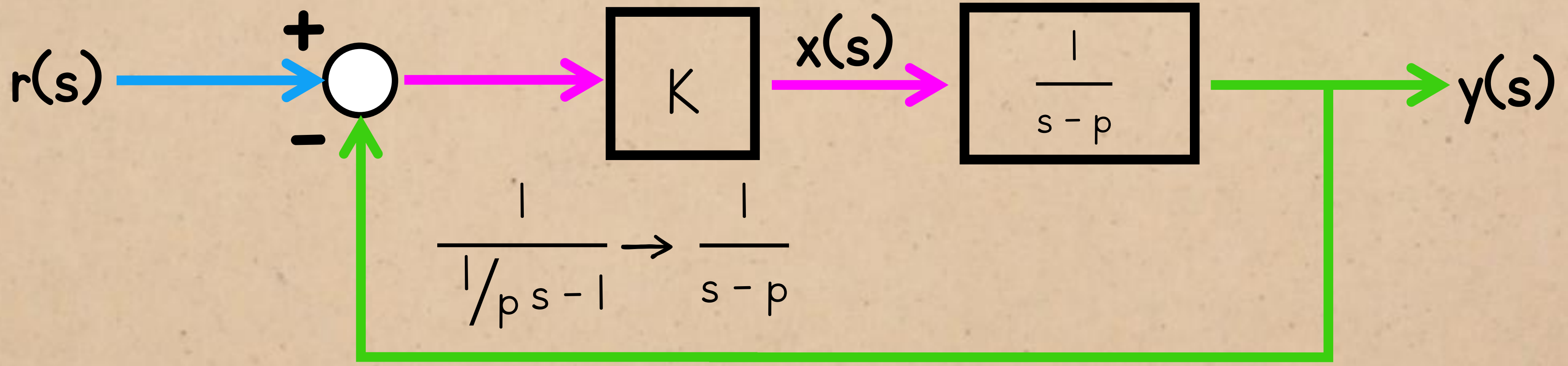
characteristic equation: $K + s - p = 0$



ROOT LOCUS

First-order system:

characteristic equation: $K + s - p = 0$



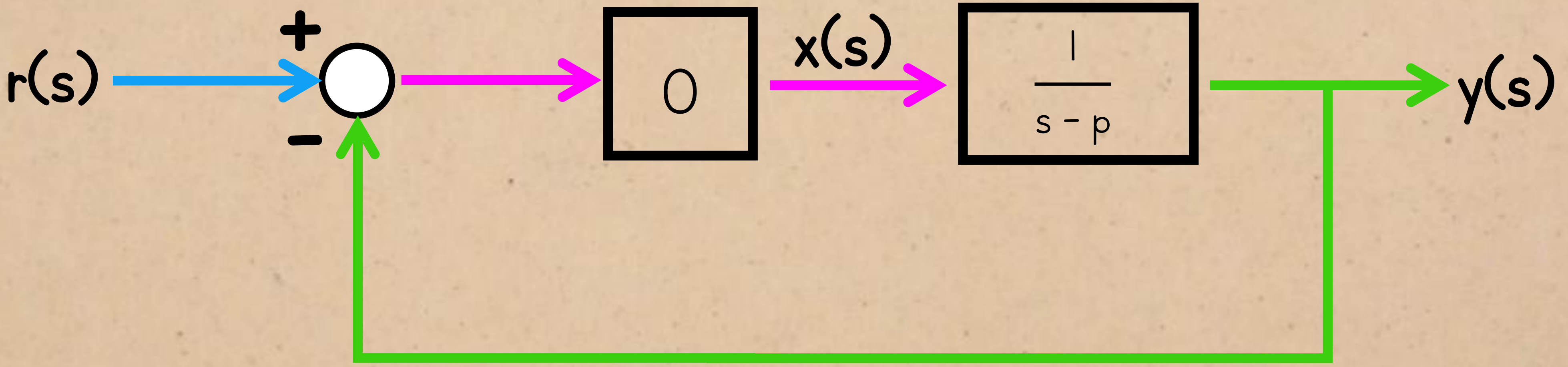
To work out the root locus, we use a version of the transfer function $Q^*(s)$ which has been scaled so the numerator and denominator are **monic** polynomials i.e. the coefficient of the highest power of s is 1.

ROOT LOCUS

When the gain = 0

characteristic equation: $s - p = 0$

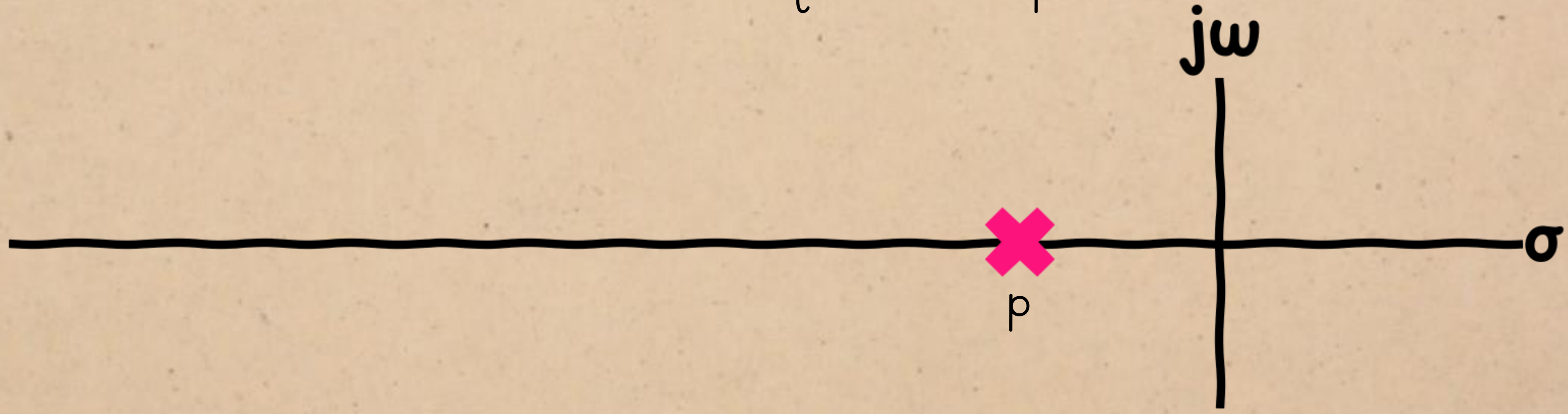
same as the open-loop characteristic equation.



ROOT LOCUS

same as the open-loop characteristic equation.

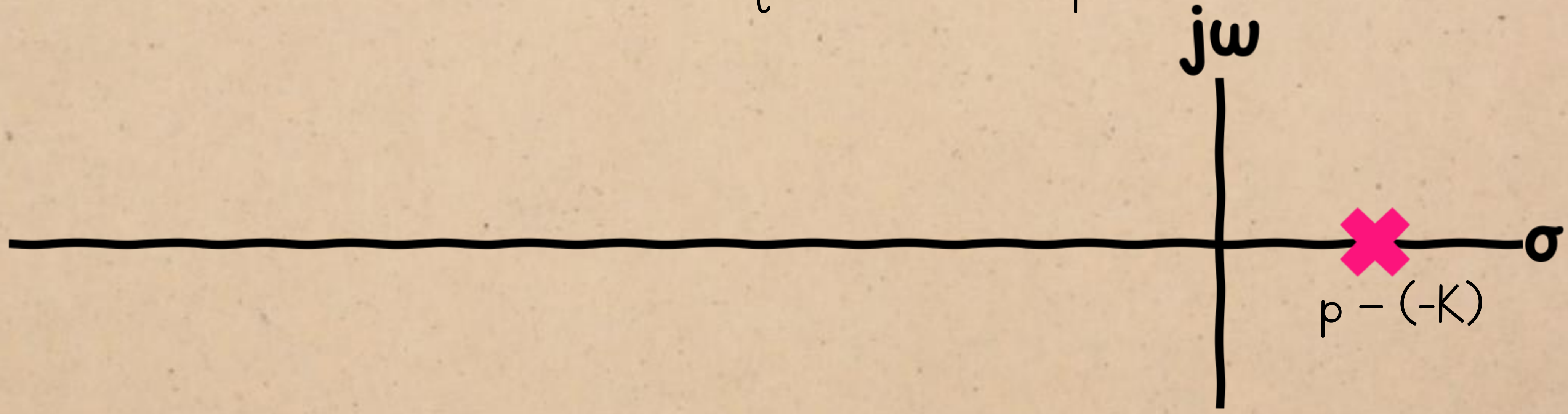
characteristic equation: $s - p = 0$



∴ the starting location of the pole will be the open-loop pole position.

ROOT LOCUS

characteristic equation: $K + s - p = 0$

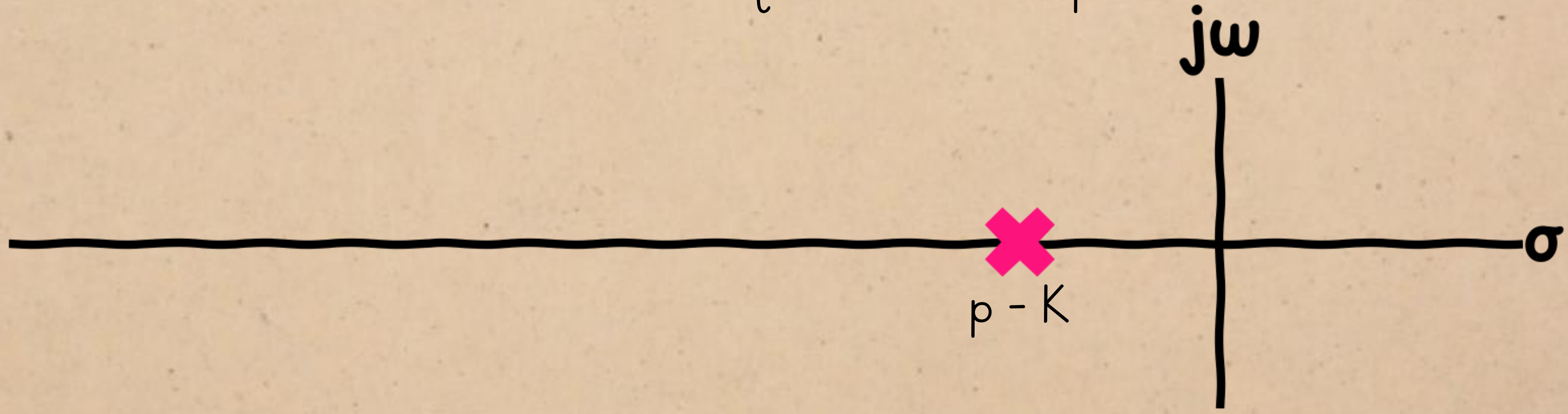


The pole can only become positive if the gain is negative

∴ the system will be stable for all the values we're considering.

ROOT LOCUS

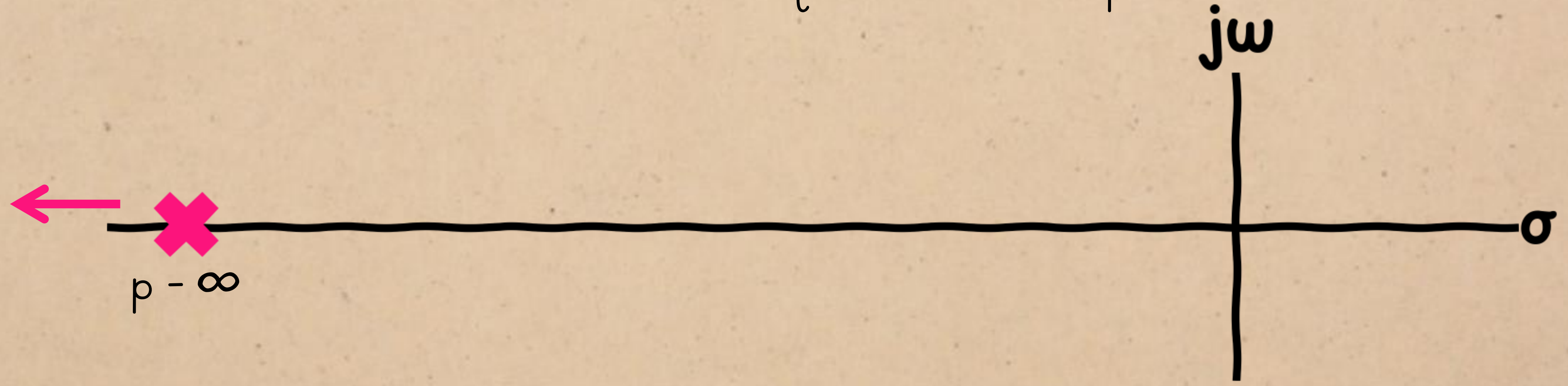
characteristic equation: $K + s - p = 0$



The pole approaches $-\infty$ as the gain increases.

ROOT LOCUS

characteristic equation: $\infty + s - p = 0$

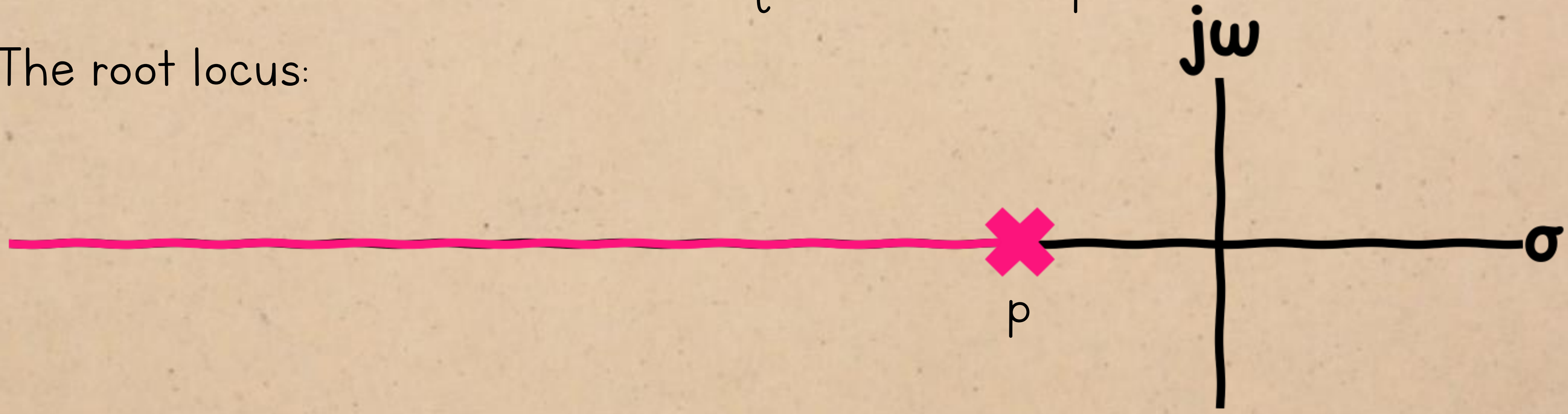


The pole approaches $-\infty$ as the gain increases.

ROOT LOCUS

characteristic equation: $K + s - p = 0$

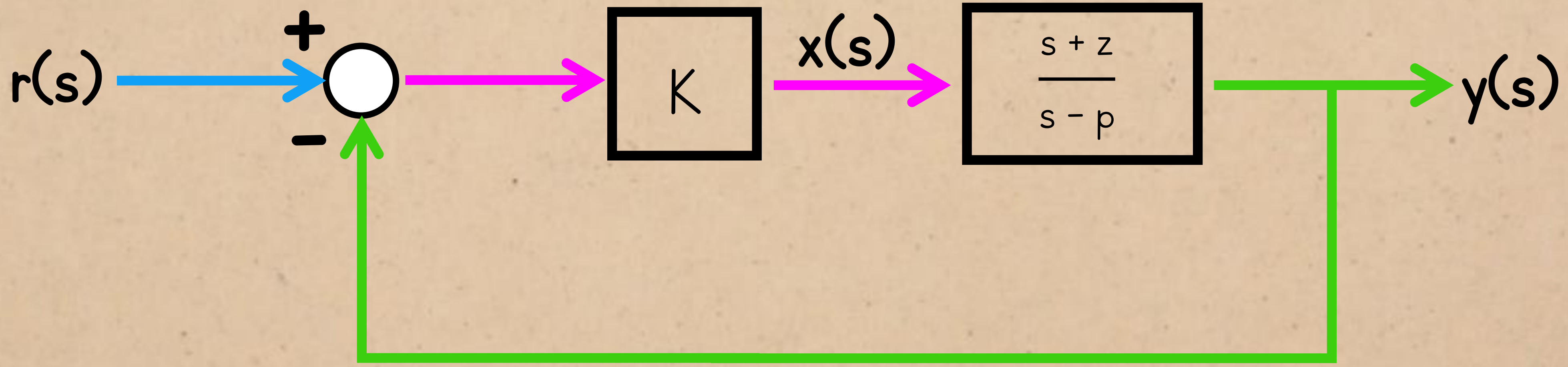
The root locus:



ROOT LOCUS

First-order system with a zero:

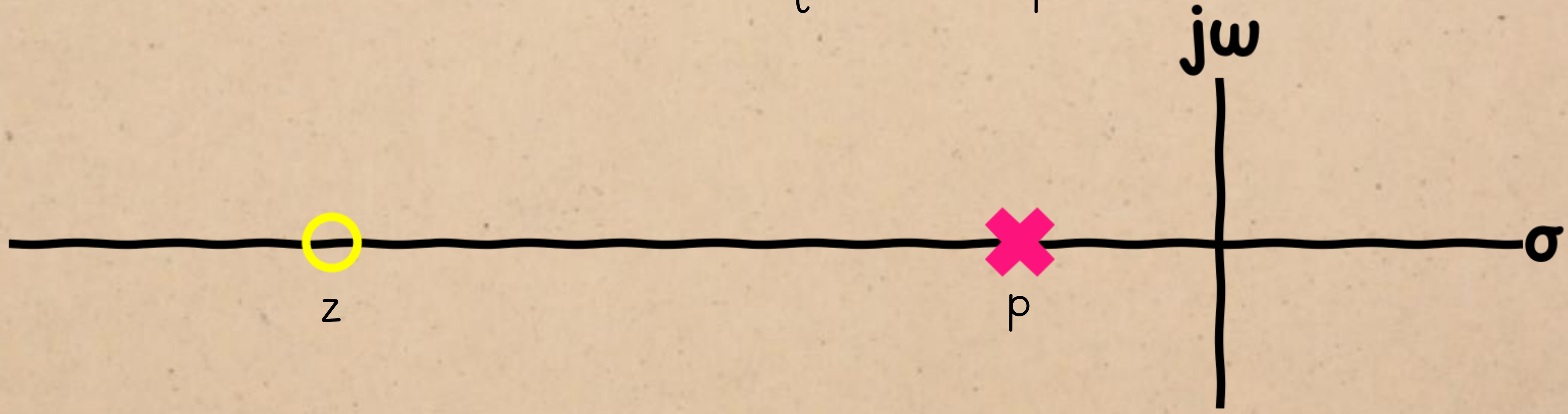
characteristic equation: $K(s + z) + s - p = 0$



ROOT LOCUS

When K = 0

characteristic equation: $s - p = 0$



∴ the starting location of the pole is still the open-loop pole position.

ROOT LOCUS

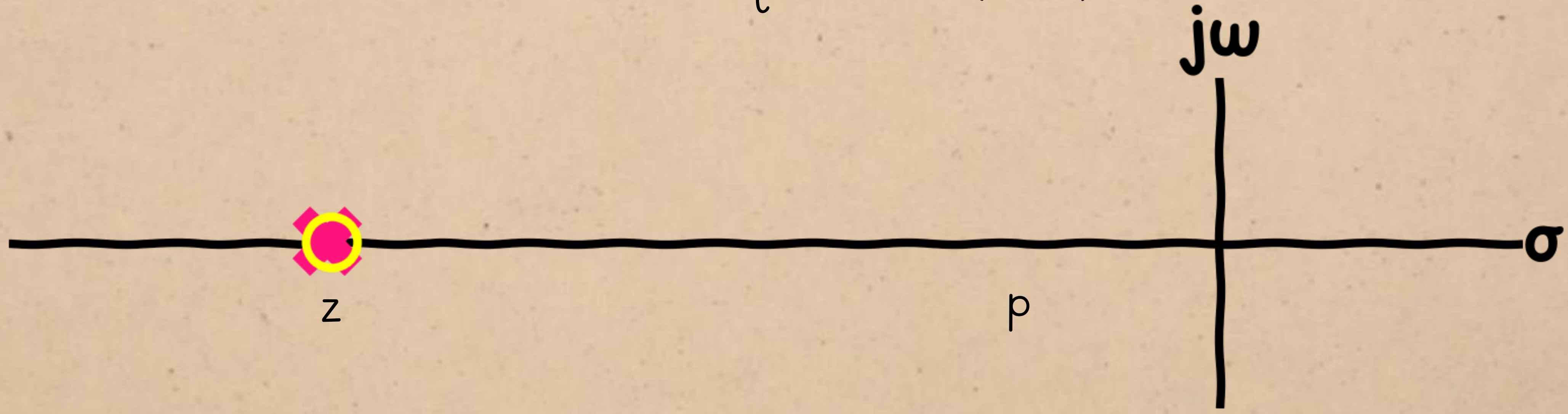
As the gain increases, the numerator term will come to dominate the characteristic equation.

characteristic equation: $\infty(s - z) + s - p = 0$

ROOT LOCUS

As the gain increases, the numerator term will come to dominate the characteristic equation.

characteristic equation: $\infty(s - z) = 0$

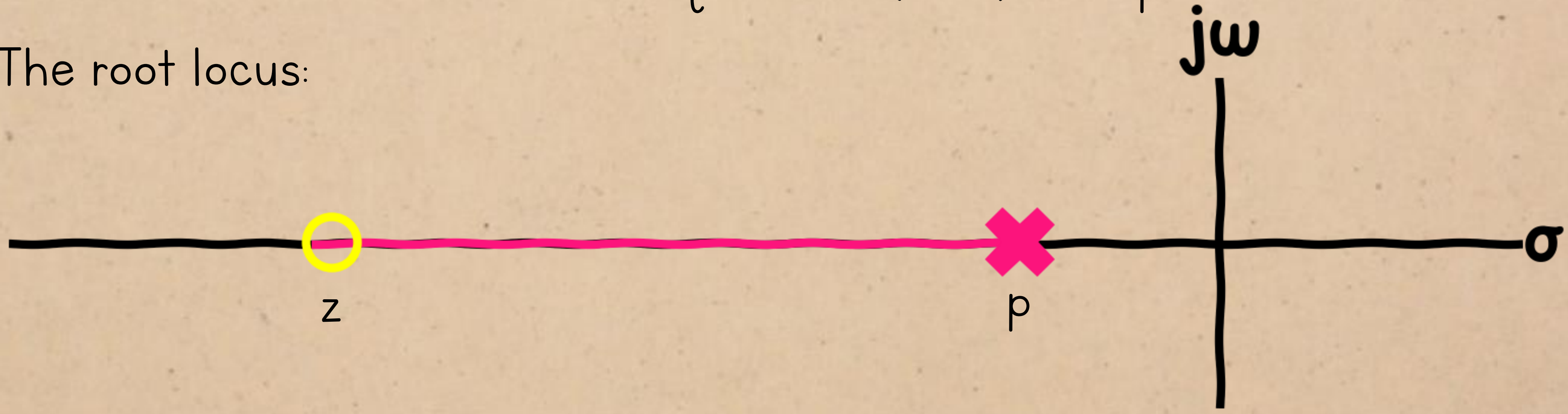


The pole approaches the zero as the gain increases.

ROOT LOCUS

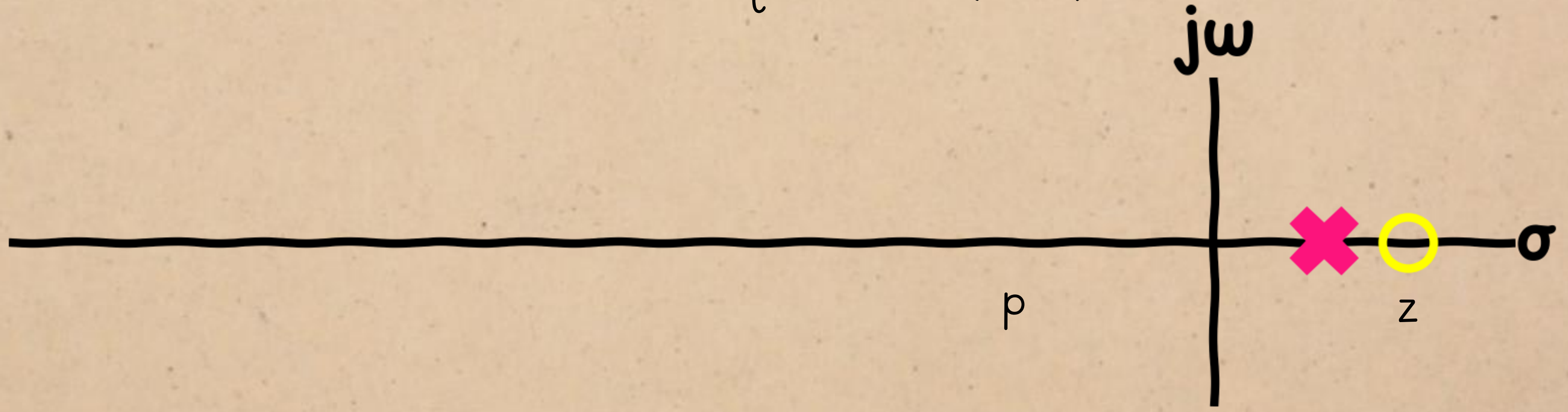
characteristic equation: $K(s - z) + s - p = 0$

The root locus:



ROOT LOCUS

characteristic equation: $\infty(s - z) = 0$



If the zero is in the right-half plane, the system could become unstable at high gains.

ROOT LOCUS

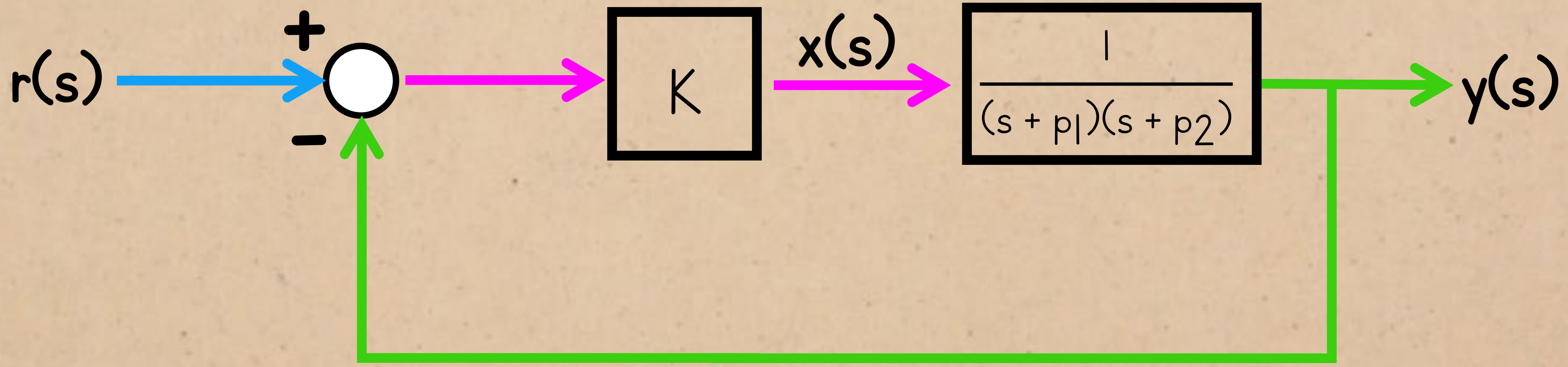
Observations

Closed-loop poles move from the open-loop poles to the open-loop zeros, or $-\infty$ if there is no zero available.

ROOT LOCUS

second-order system:

characteristic equation: $K + (s + p_1)(s + p_2) = 0$

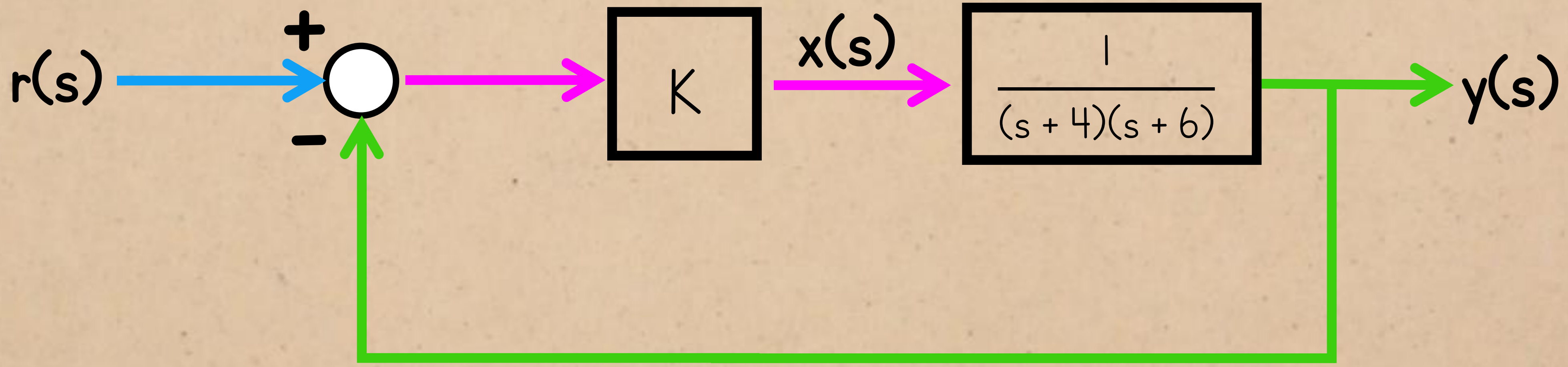


(Using the system from the proportional control example, since we've already seen where the closed-loop poles end up for some gains.)

ROOT LOCUS

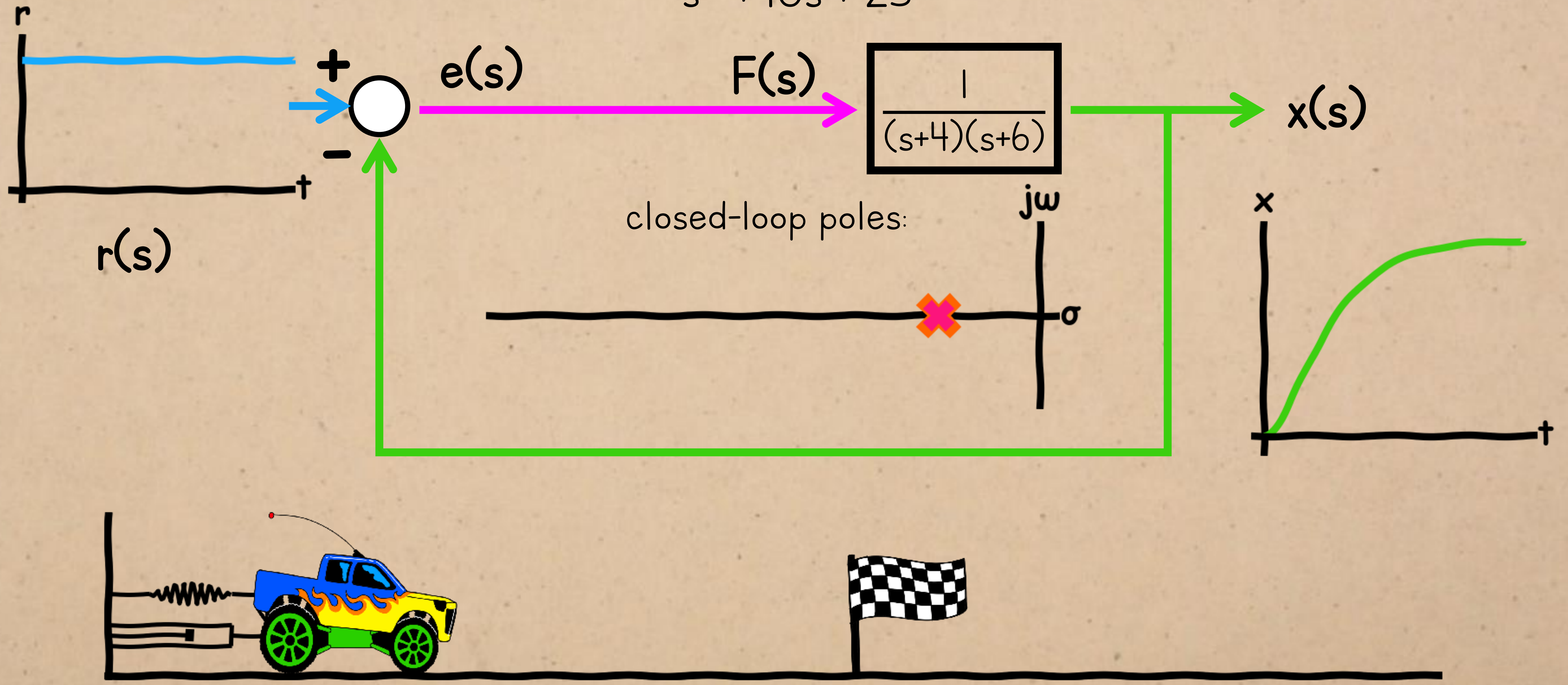
second-order system:

characteristic equation: $K + (s + 4)(s + 6) = 0$



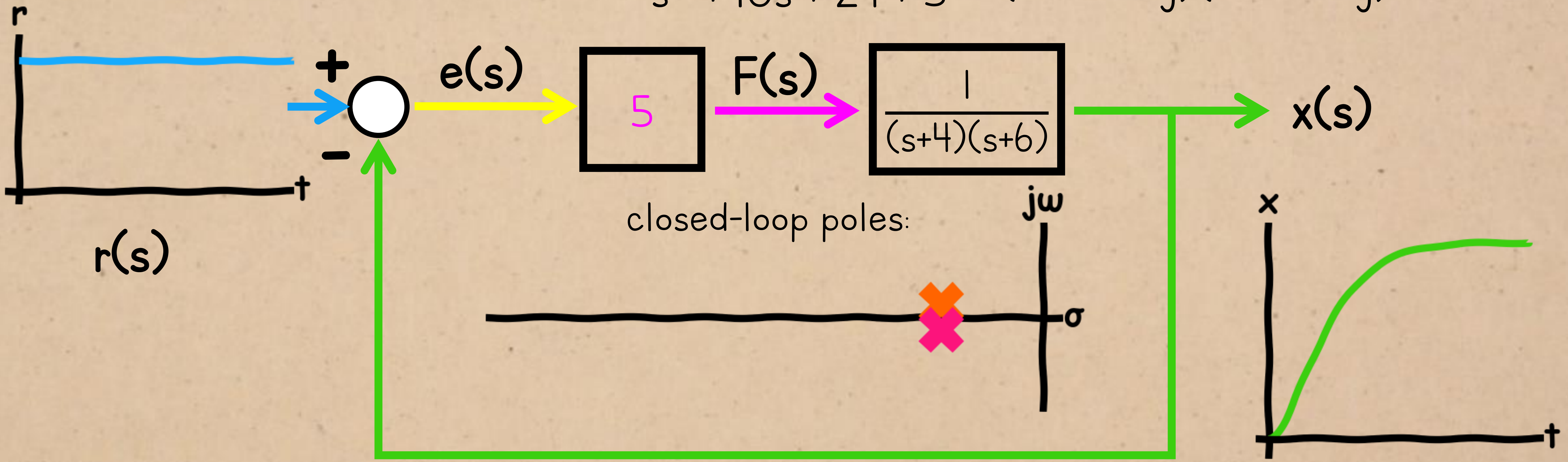
LAST TIME:

$$G_{CL} = \frac{1}{s^2 + 10s + 25}$$



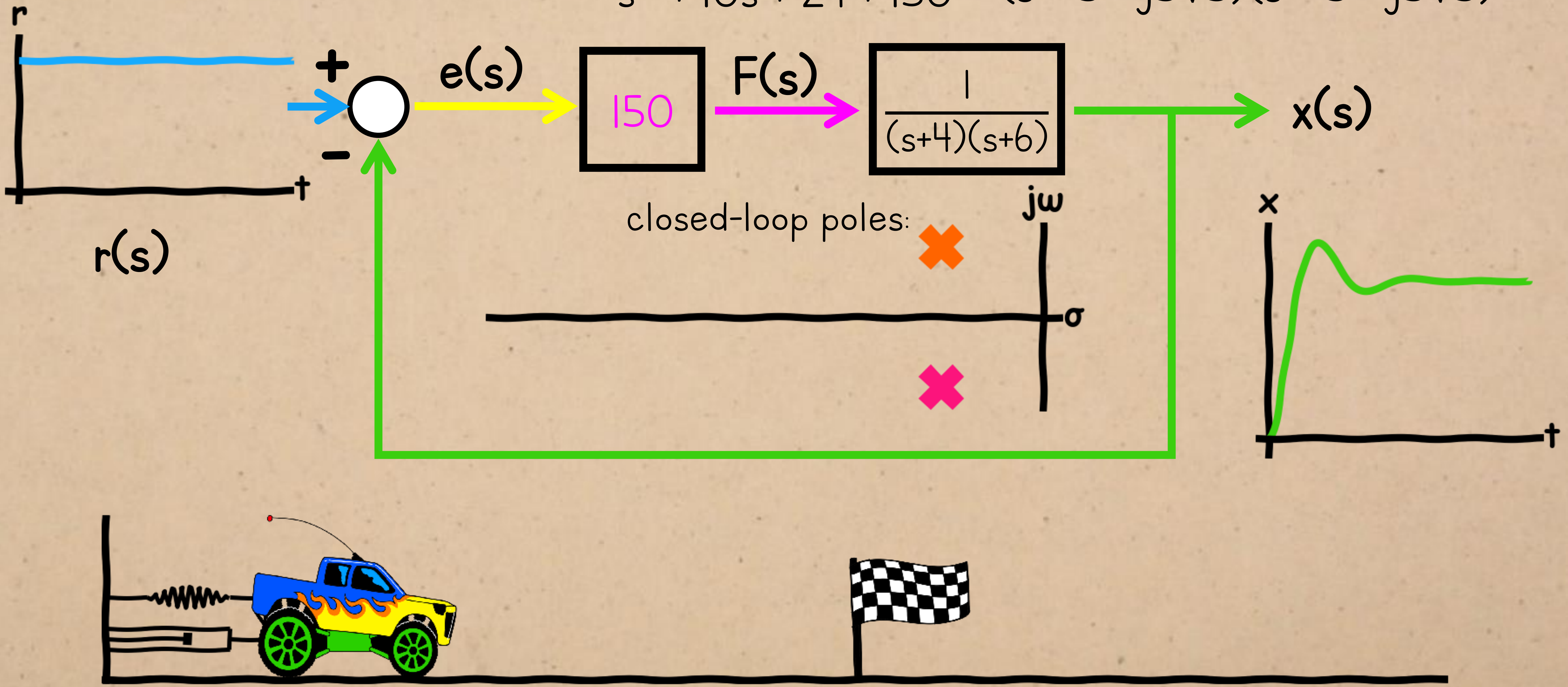
LAST TIME:

$$G_{CL} = \frac{1}{s^2 + 10s + 24 + 5} = \frac{1}{(s + 5 - 2j)(s + 5 + 2j)}$$



LAST TIME:

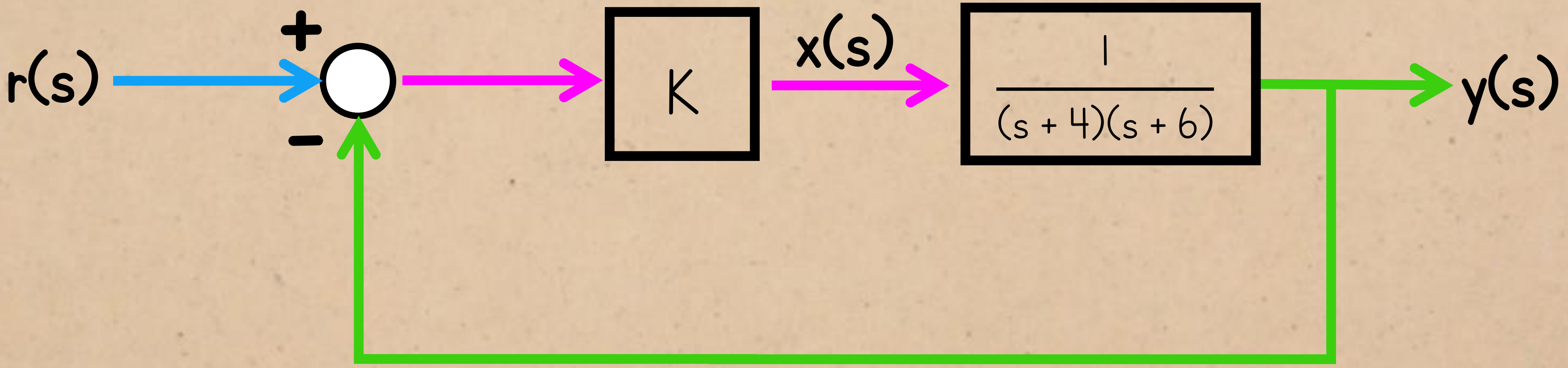
$$G_{CL} = \frac{1}{s^2 + 10s + 24 + 150} = \frac{1}{(s + 5 - j5\sqrt{5})(s + 5 + j5\sqrt{5})}$$



ROOT LOCUS

second-order system:

characteristic equation: $s^2 + 10s + (24 + K) = 0$



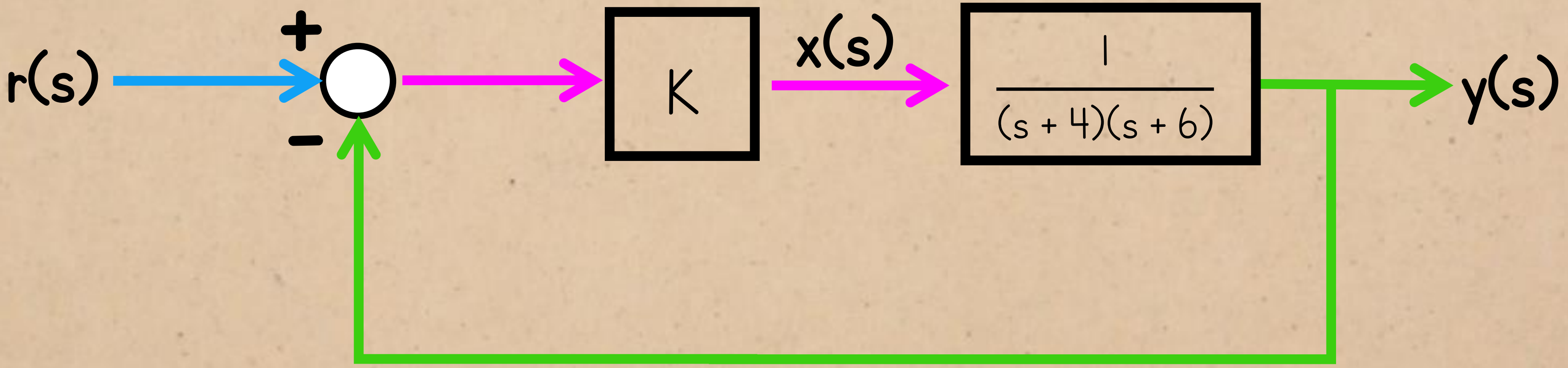
poles:

$$-5 \pm \frac{1}{2}\sqrt{10^2 - 4(24 + K)}$$

ROOT LOCUS

second-order system:

characteristic equation: $s^2 + 10s + (24 + K) = 0$

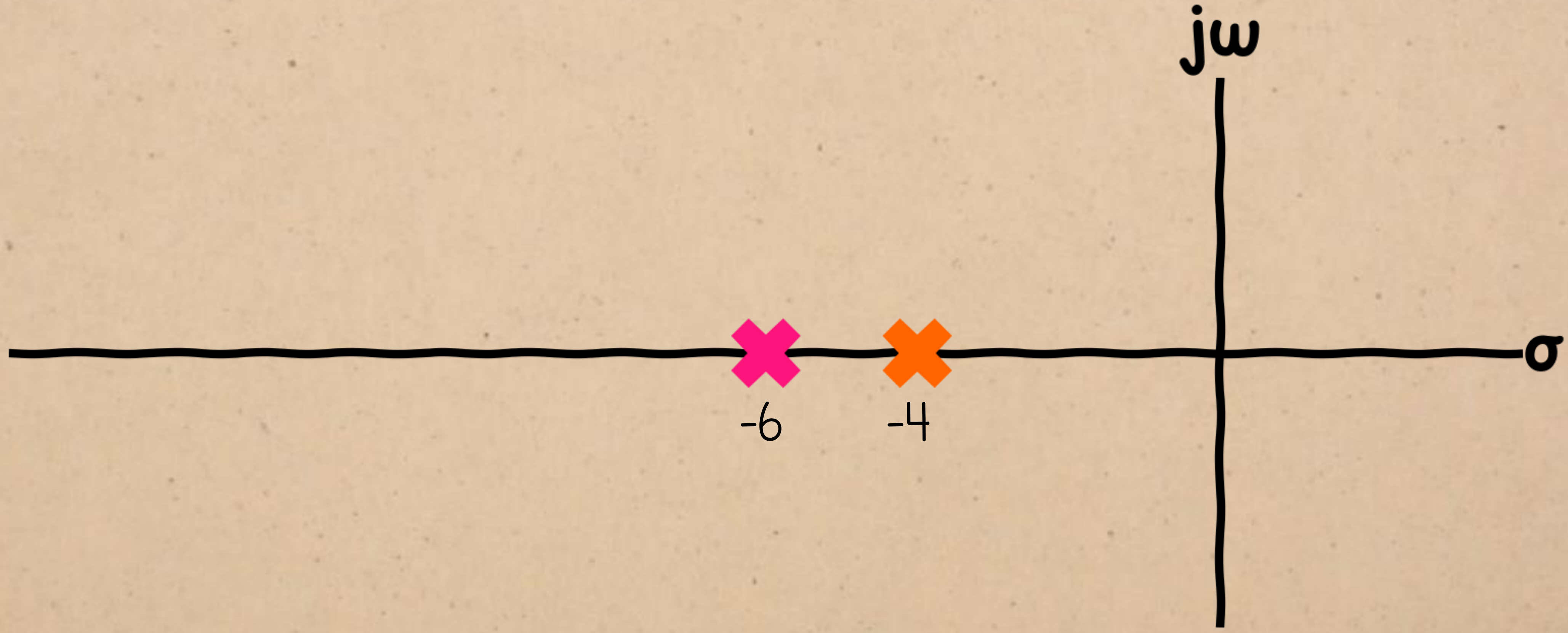


poles:

$$-5 \pm \frac{1}{2}\sqrt{10^2 - 4(24 + K)}$$

decreases as K increases

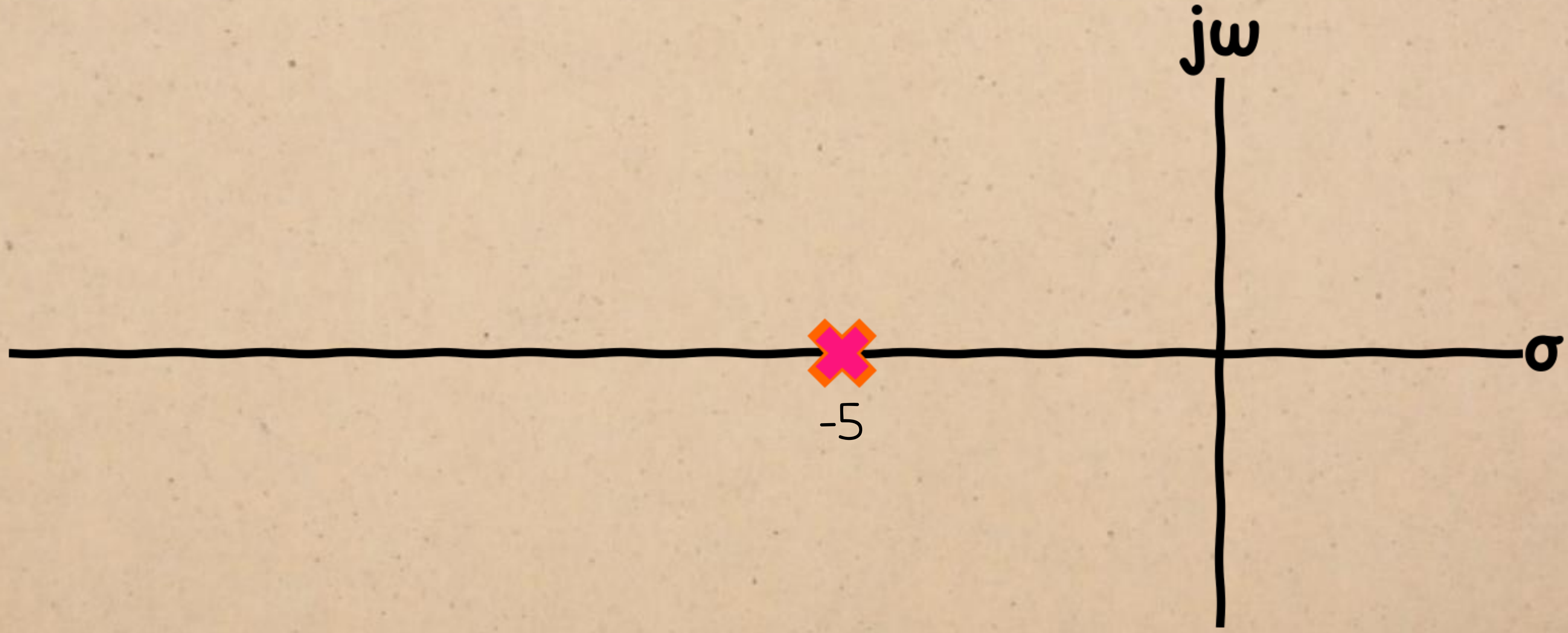
ROOT LOCUS



$$-5 \pm \frac{1}{2}\sqrt{10^2 - 4(24 + K)}$$

...so the poles go from real...

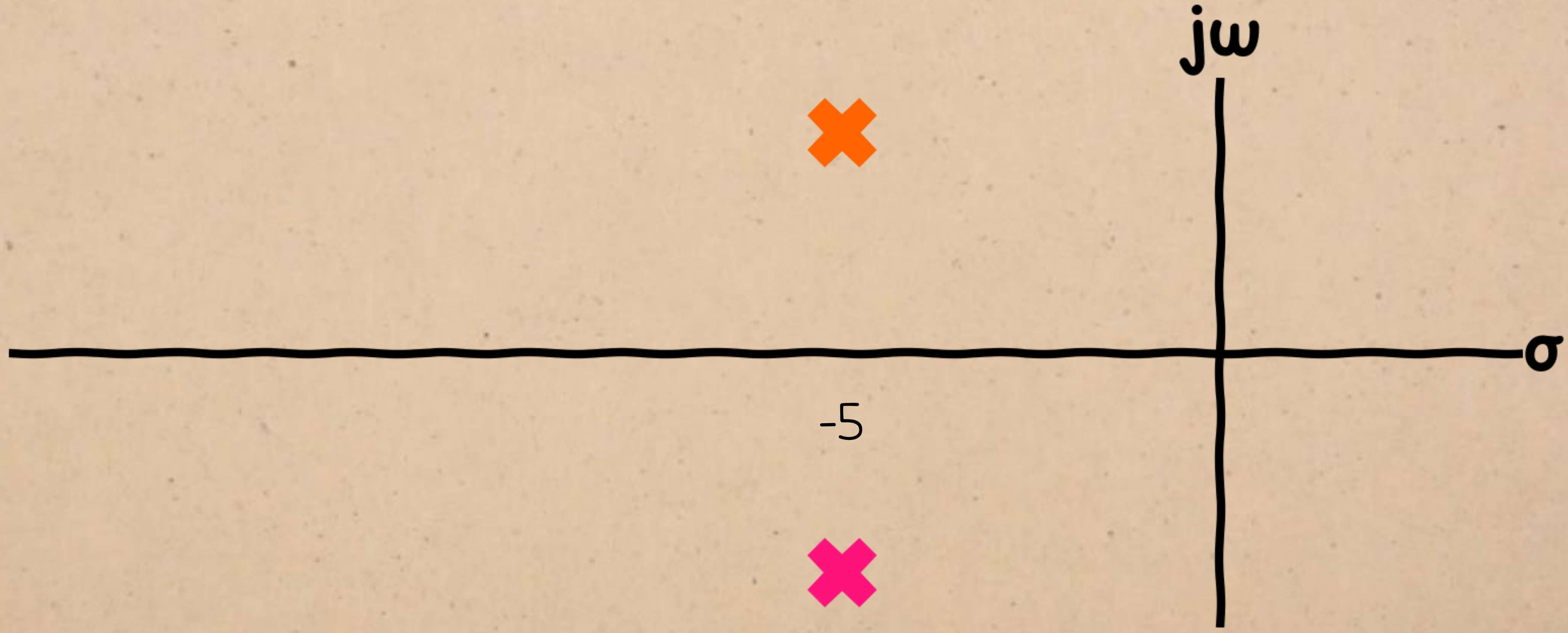
ROOT LOCUS



$$-5 \pm \frac{1}{2}\sqrt{10^2 - 4(24 + K)}$$

...so the poles go from real → repeating...

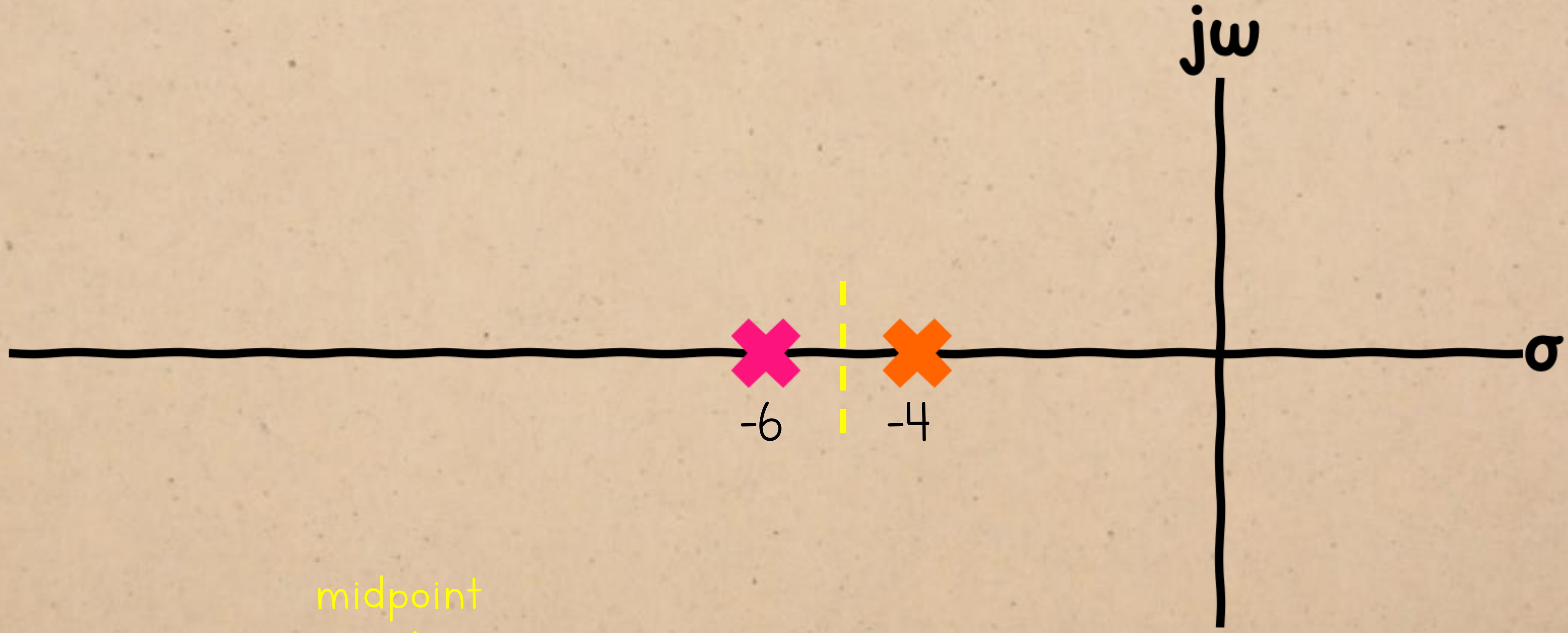
ROOT LOCUS



$$-5 \pm \frac{1}{2}\sqrt{10^2 - 4(24 + K)}$$

...so the poles go from real → repeating → complex

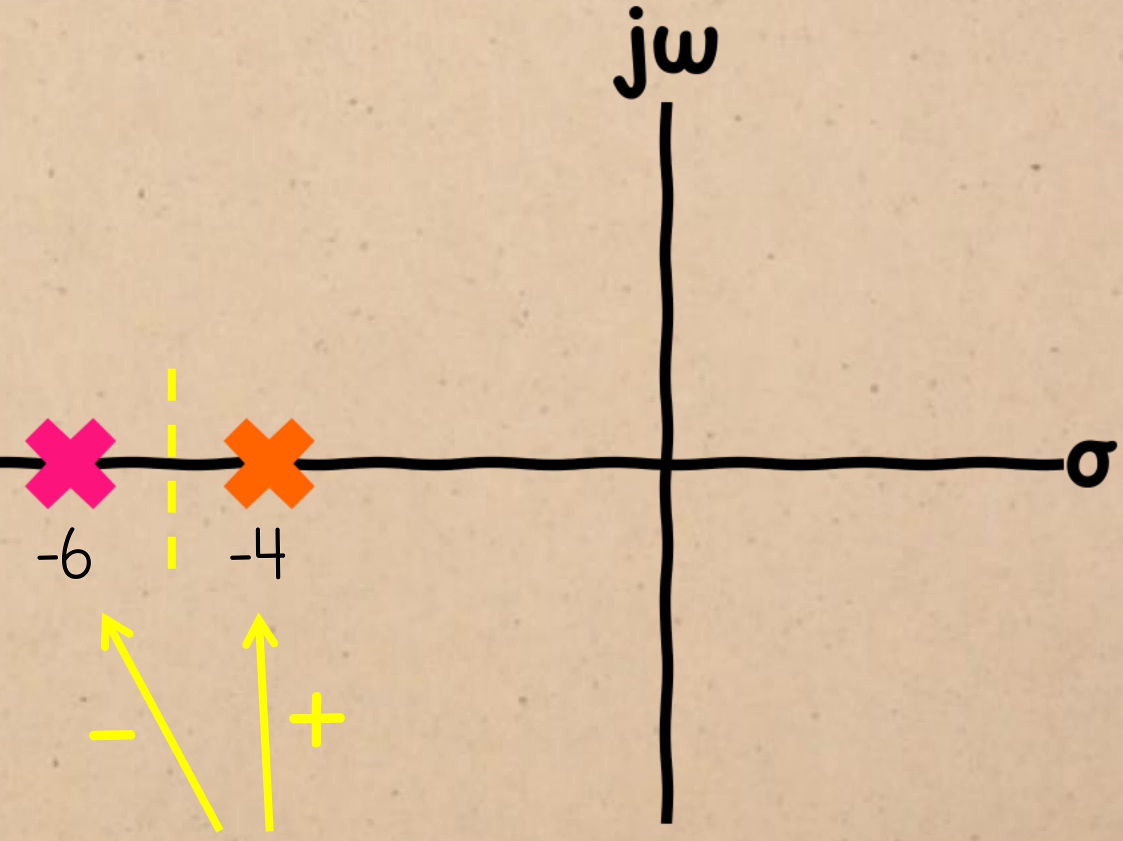
ROOT LOCUS



midpoint

$$-5 \pm \frac{1}{2}\sqrt{10^2 - 4(24 + K)}$$

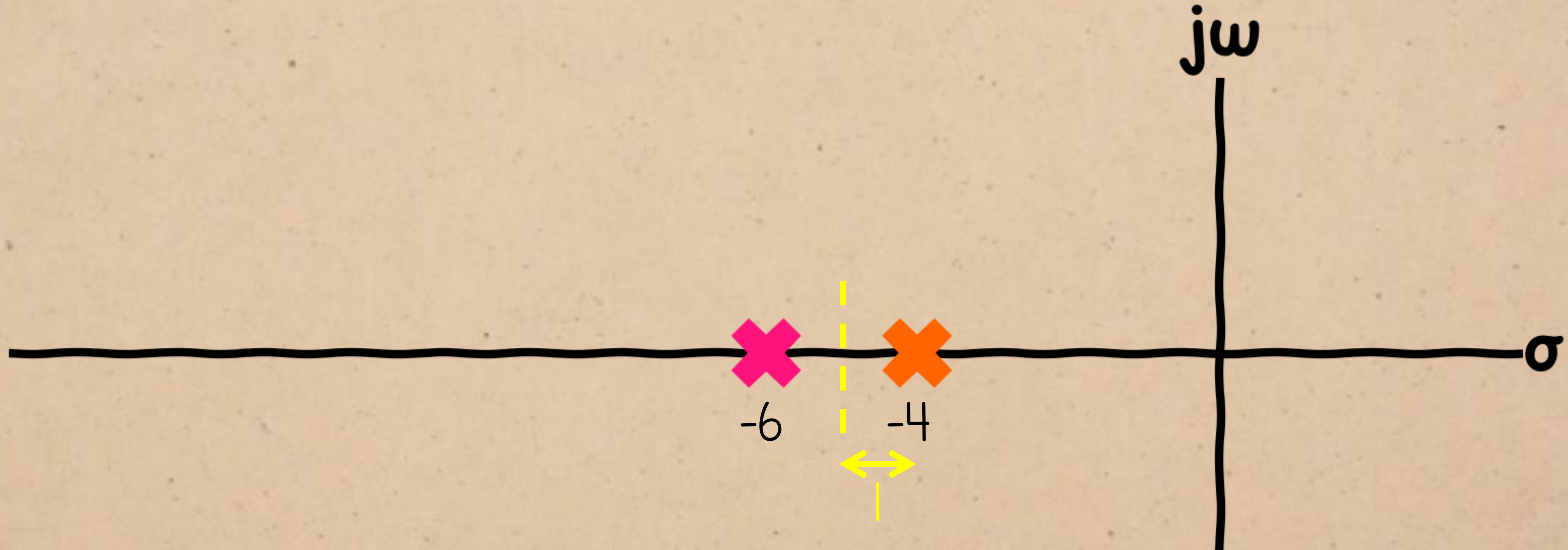
ROOT LOCUS



The square root is added to the midpoint to get the pole on the right and subtracted from it to get the pole on the left...

$$-5 \pm \frac{1}{2}\sqrt{10^2 - 4(24 + K)}$$

ROOT LOCUS



So,
The only way you'll get a pole that's < -6 or > -4 is if this is > 1.

$$-5 \pm \frac{1}{2}\sqrt{10^2 - 4(24 + K)}$$

(i.e. half the distance between the poles)

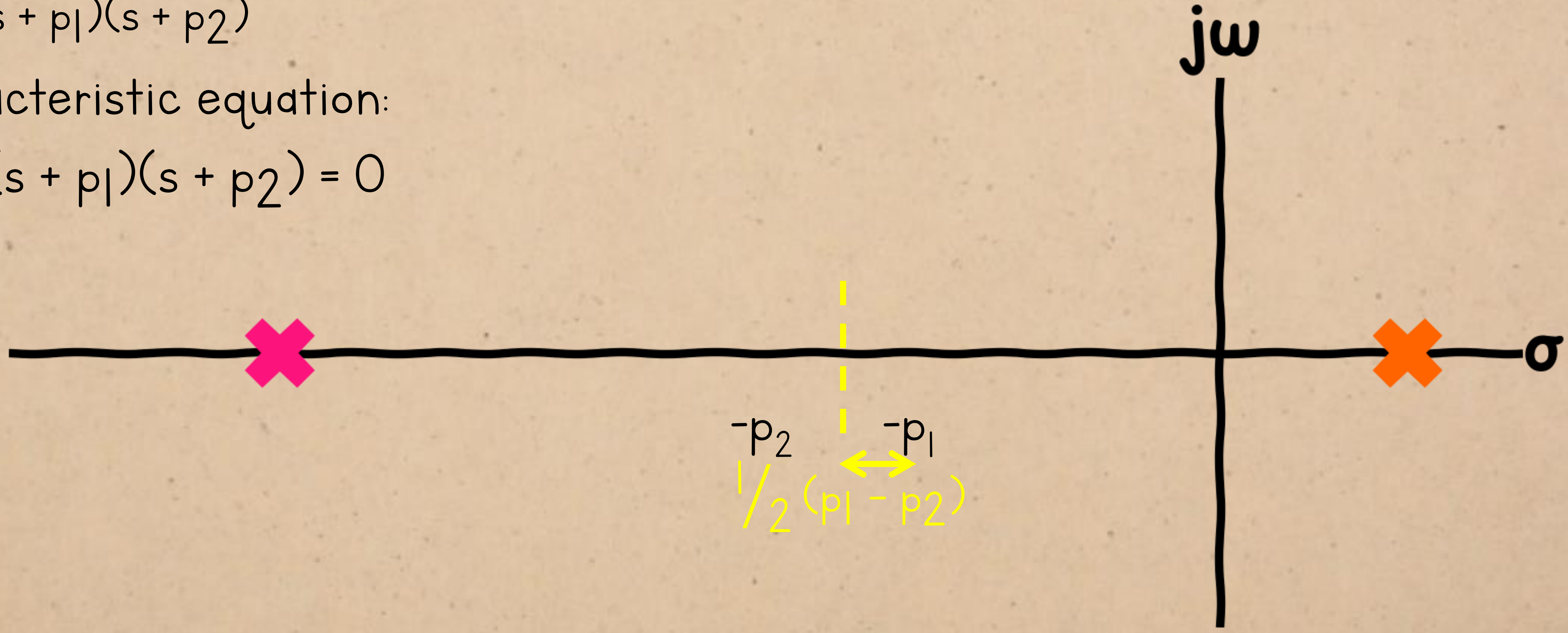
ROOT LOCUS

Checking for a generic system:

$$\frac{1}{(s + p_1)(s + p_2)}$$

characteristic equation:

$$K + (s + p_1)(s + p_2) = 0$$



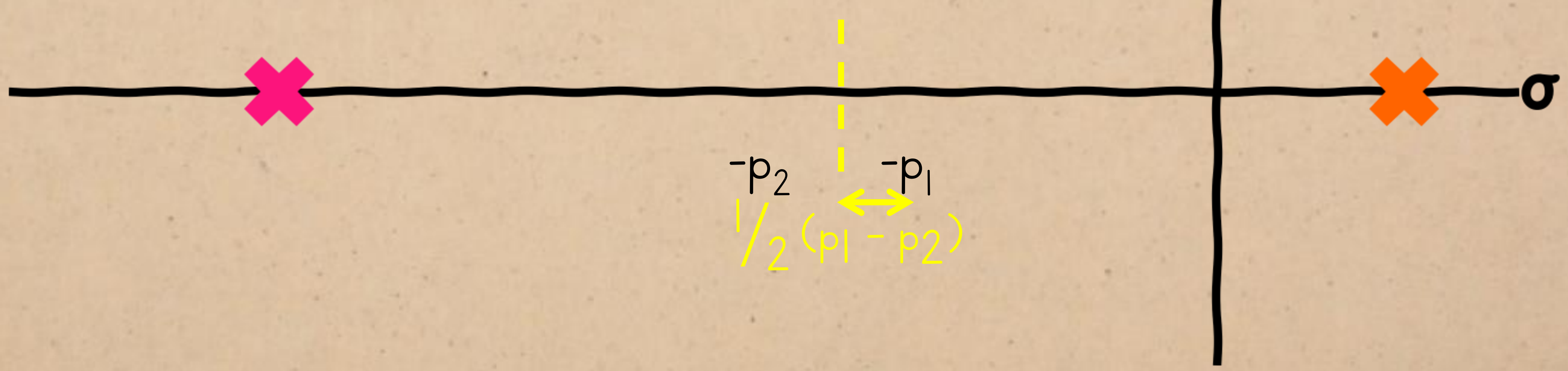
$$-\frac{p_1 + p_2}{2} \pm \frac{1}{2}\sqrt{(p_1 + p_2)^2 - 4(p_1 p_2 + K)}$$

ROOT LOCUS

$$\frac{1}{(s+p_1)(s+p_2)}$$

characteristic equation:

$$K + (s+p_1)(s+p_2) = 0$$



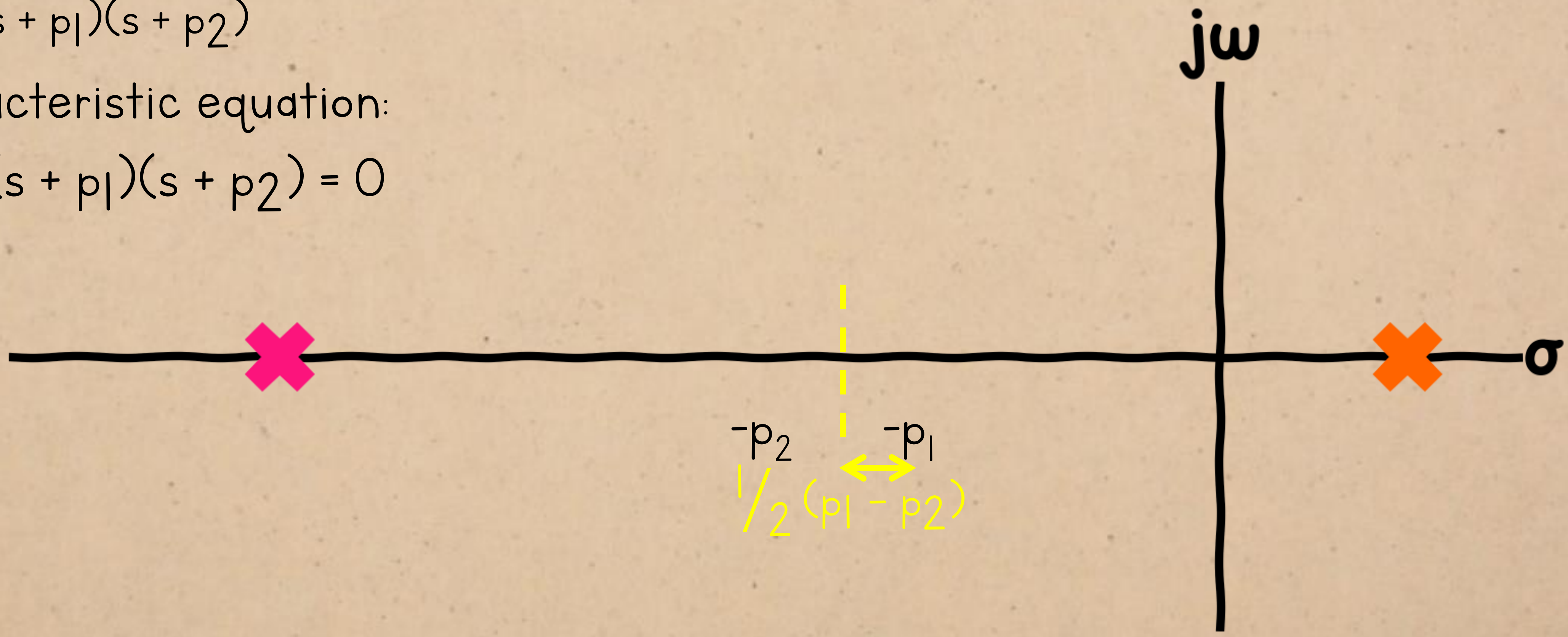
$$-\frac{p_1+p_2}{2} \pm \frac{1}{2}\sqrt{(p_1 - p_2)^2 - 4K}$$

ROOT LOCUS

$$\frac{1}{(s + p_1)(s + p_2)}$$

characteristic equation:

$$K + (s + p_1)(s + p_2) = 0$$

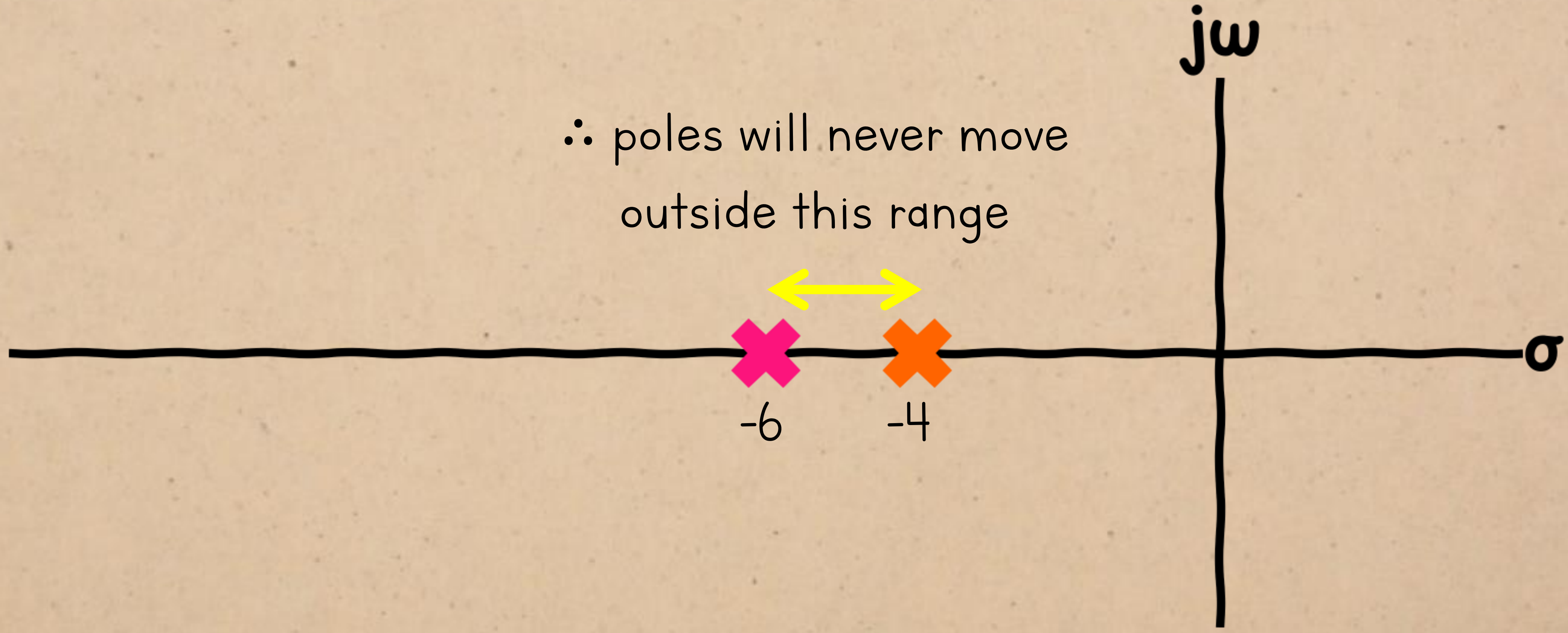


$$-\frac{p_1 + p_2}{2} \pm \frac{1}{2}\sqrt{(p_1 - p_2)^2 - 4K}$$

∴ only possible if the gain is negative.

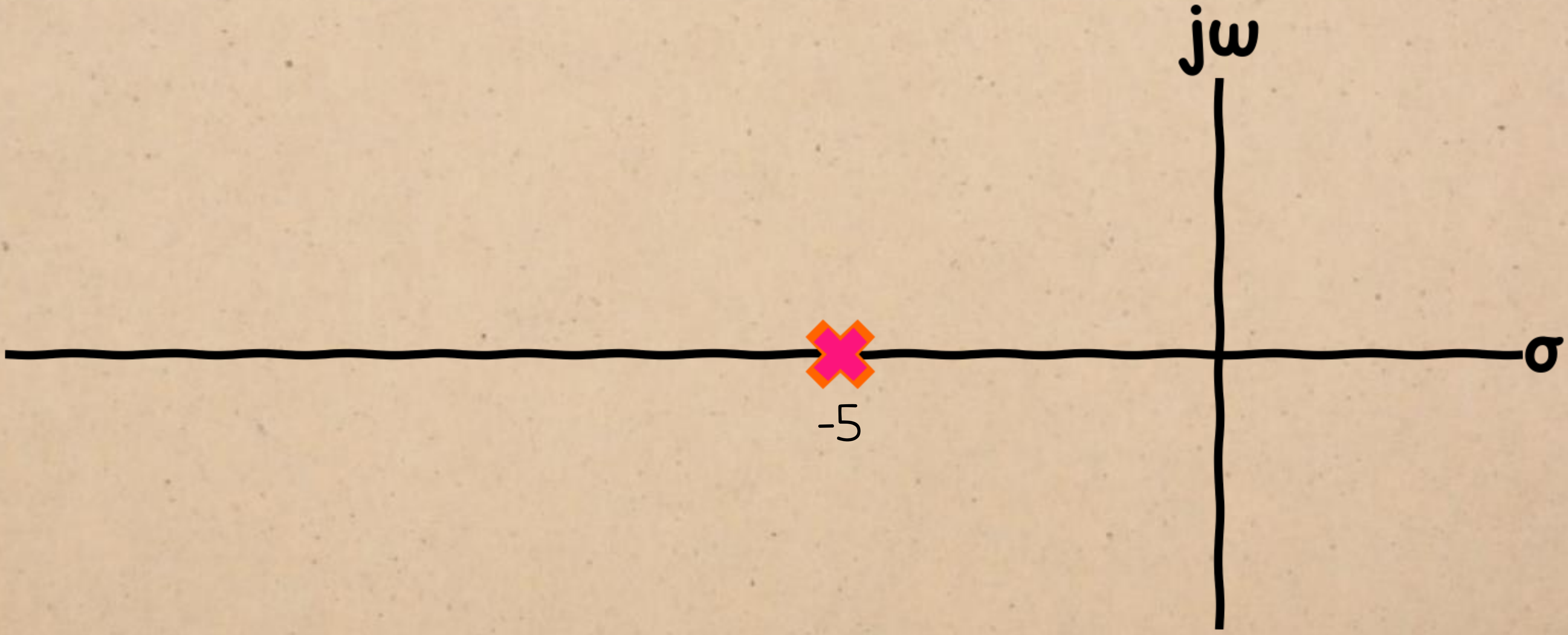
ROOT LOCUS

∴ poles will never move outside this range



$$-5 \pm \frac{1}{2}\sqrt{10^2 - 4(24 + K)}$$

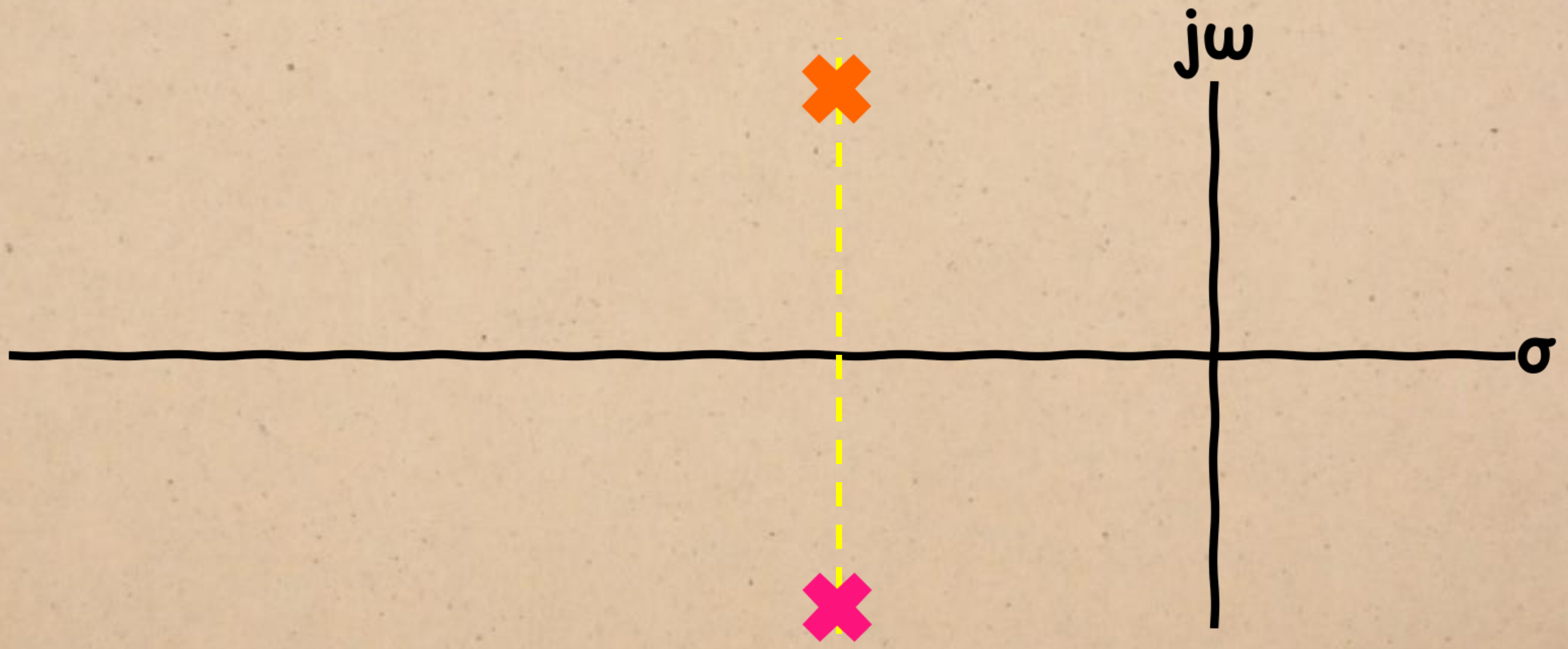
ROOT LOCUS



$$-5 \pm \frac{1}{2}\sqrt{10^2 - 4(24 + K)}$$

if this is ≤ 0, real parts = -5

ROOT LOCUS

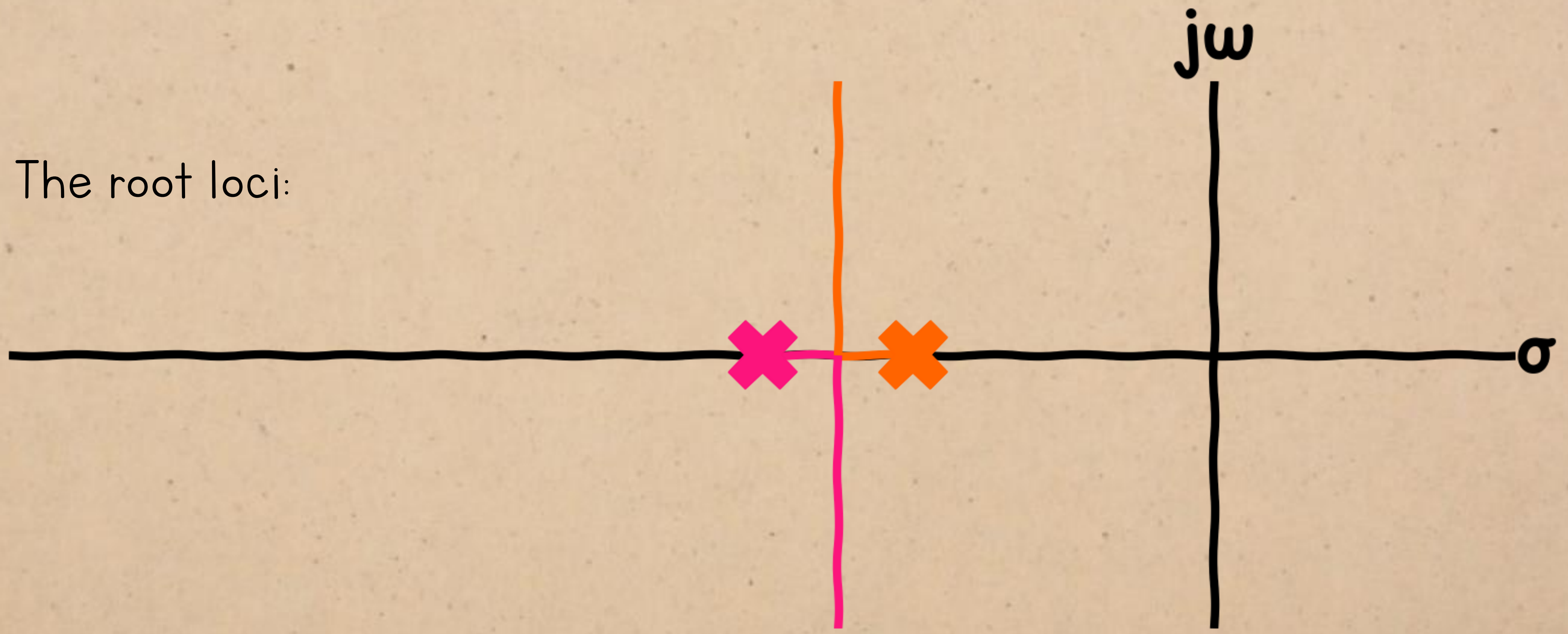


$$-5 \pm \frac{1}{2}\sqrt{10^2 - 4(24 + K)}$$

the imaginary parts tend to $\pm\infty$ as the gain increases

ROOT LOCUS

The root loci:



ROOT LOCUS

For complex open-loop poles...

The root loci:

the closed-loop poles will always be complex.

ROOT LOCUS

For complex open-loop poles...

The root loci:

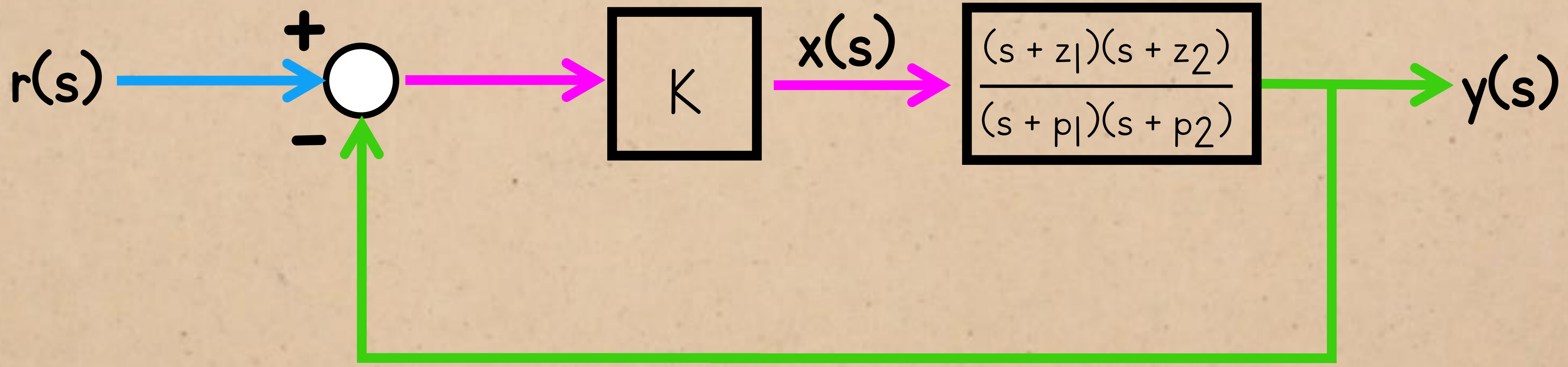
$$-\frac{p_1+p_2}{2} \pm \frac{1}{2}\sqrt{(p_1-p_2)^2-4K}$$

...because this value will never be larger than it is for the open-loop poles.

ROOT LOCUS

second-order system with two zeros:

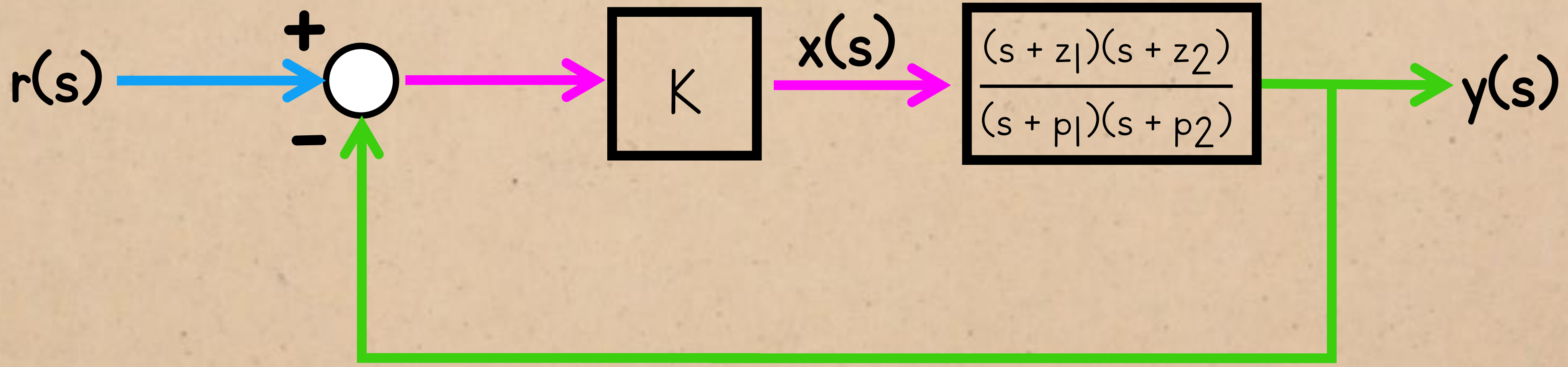
characteristic equation: $K(s + z_1)(s + z_2) + (s + p_1)(s + p_2) = 0$



ROOT LOCUS

second-order system with two zeros:

characteristic equation: $\infty(s + z_1)(s + z_2) + (s + p_1)(s + p_2) = 0$



ROOT LOCUS

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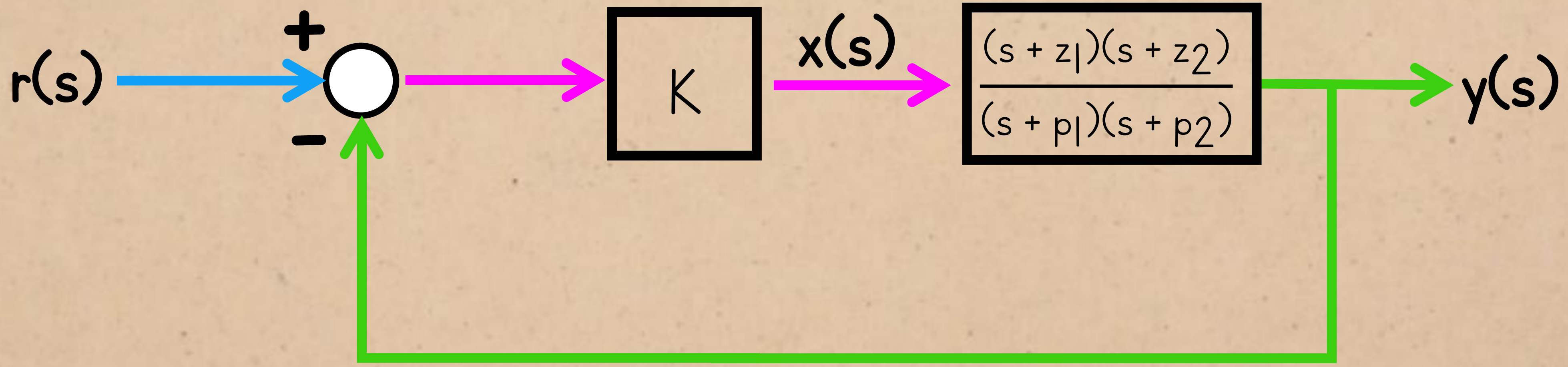
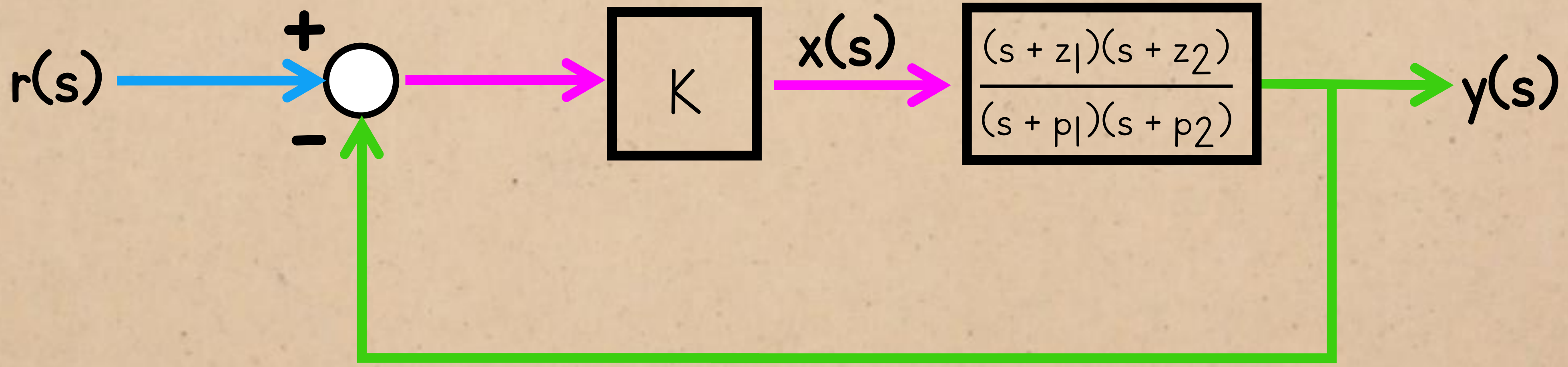


ROOT LOCUS

second-order system with two zeros:

characteristic equation: $\infty(s + z_1)(s + z_2) = 0$

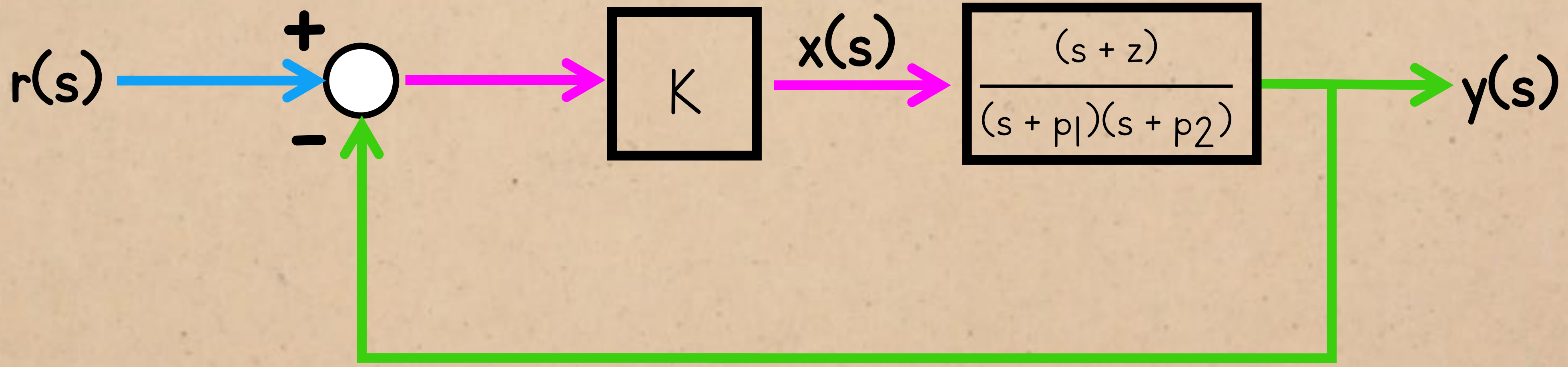


$\therefore$ the closed-loop poles tend to the zero positions

ROOT LOCUS

second-order system with one zero:

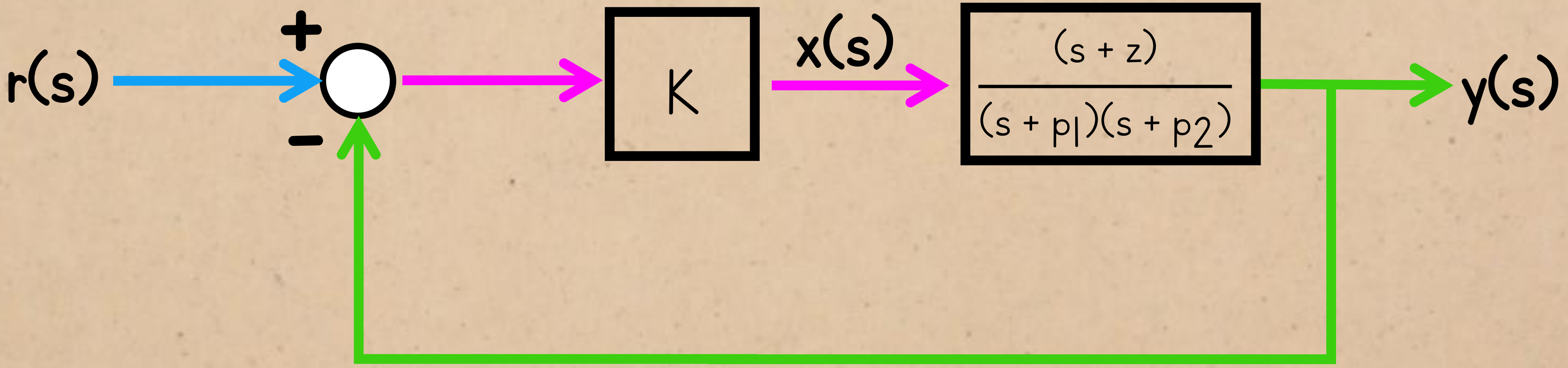
characteristic equation: $K(s + z) + (s + p_1)(s + p_2) = 0$



ROOT LOCUS

second-order system with one zero:

characteristic equation: $\infty(s + z) + (s + p_1)(s + p_2) = 0$



ROOT LOCUS

second-order system with one zero:

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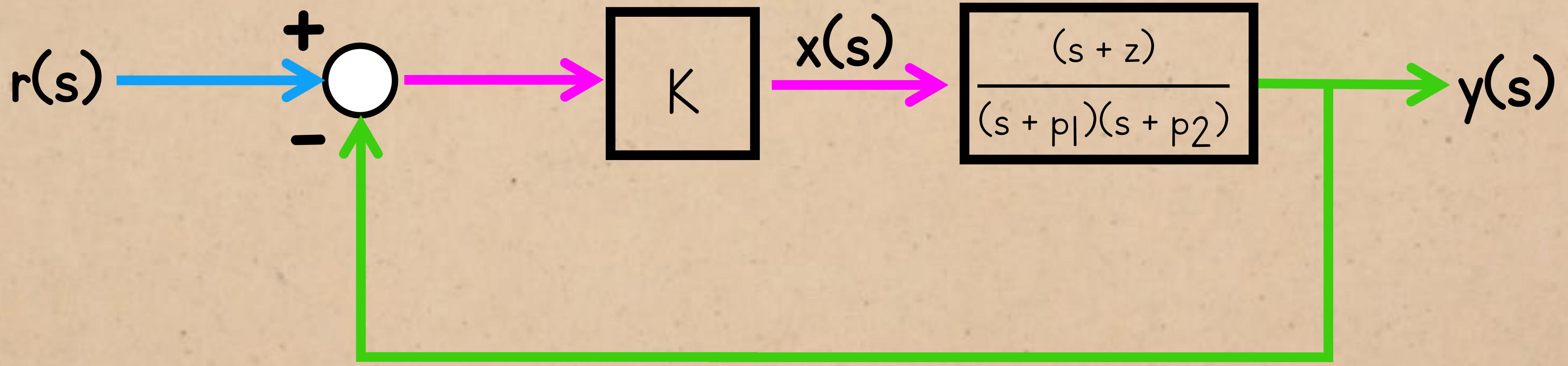

∴ one pole tends to z

and the other tends to $-\infty$

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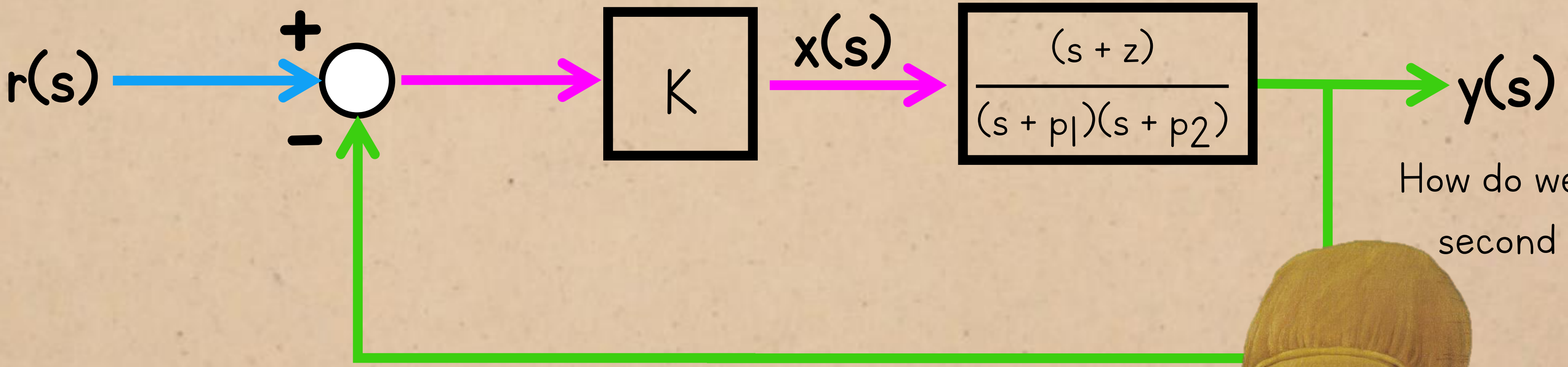
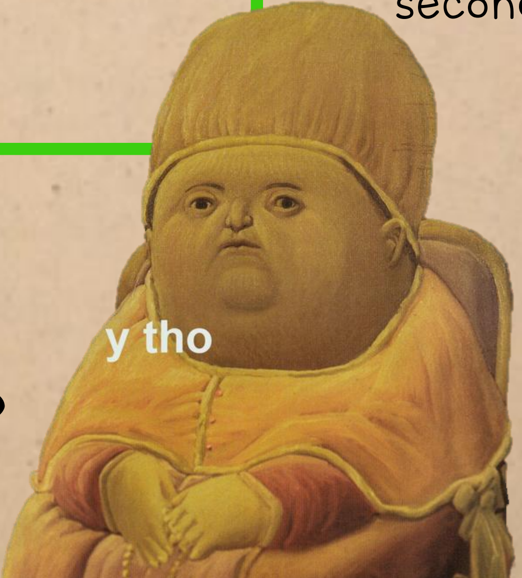
second-order system with one zero:

characteristic equation: $\infty(s + z) = 0$



How do we know what the second pole is doing?



$\therefore$ one pole tends to z

and the other tends to $-\infty$

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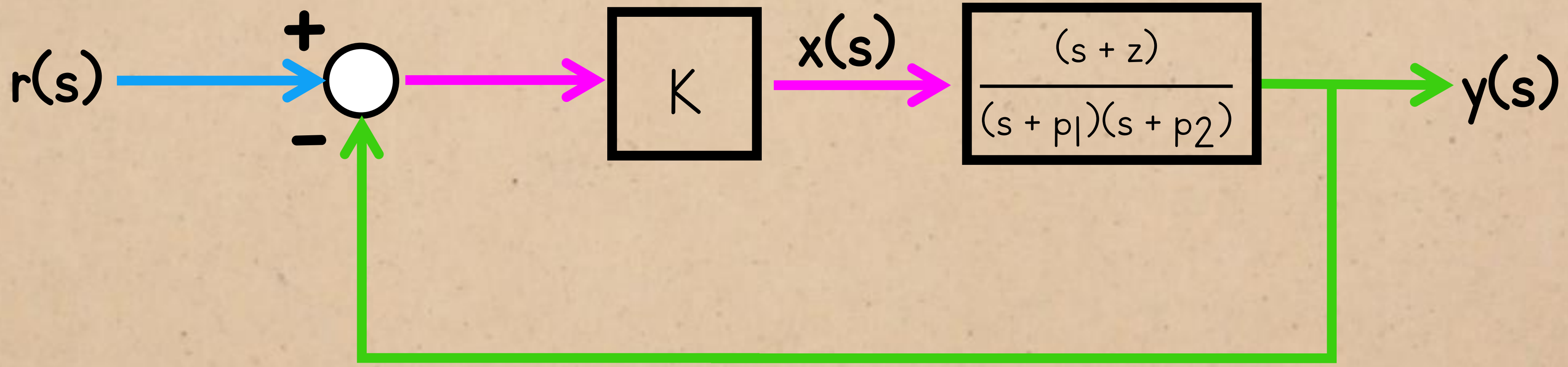
This is what the characteristic polynomial's factors would have to be if that's true:

second-order system with one zero:

characteristic equation: $(s + z)(s + \infty) = 0$



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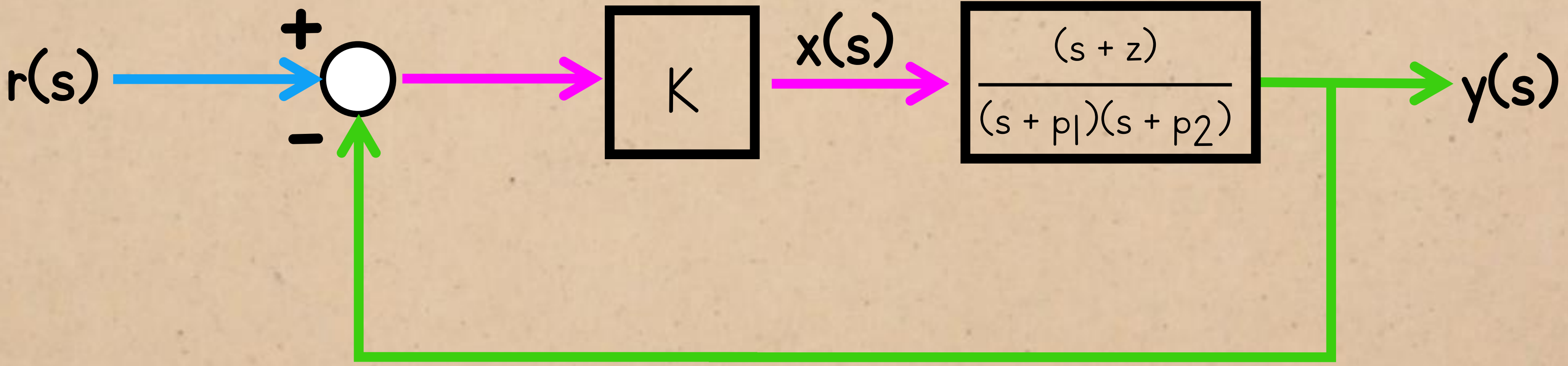
second-order system with one zero:

characteristic equation: $s^2 + (\infty + z)s + \infty z$



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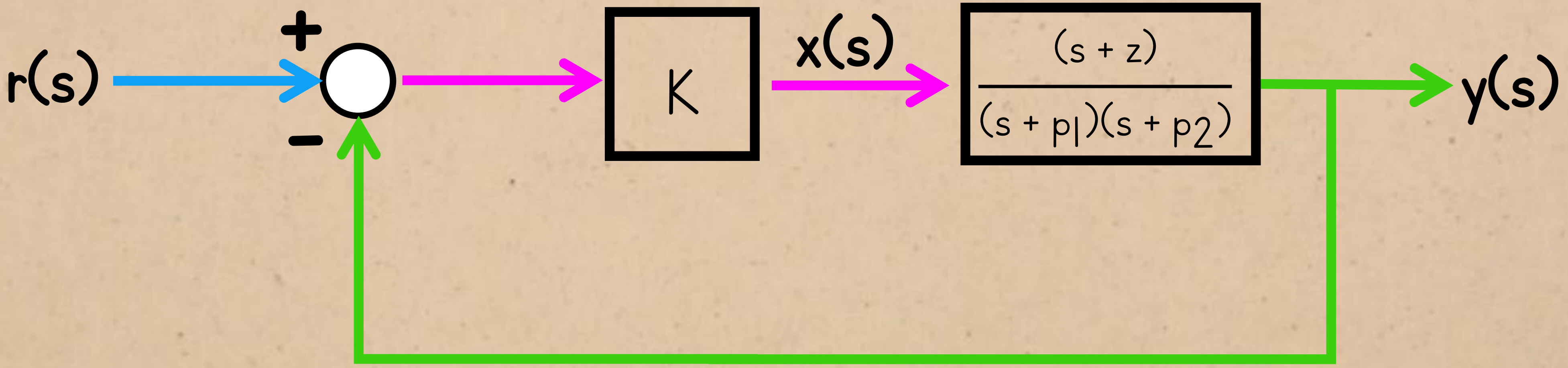
second-order system with one zero:

characteristic equation: $s^2 + \infty s + \infty z$



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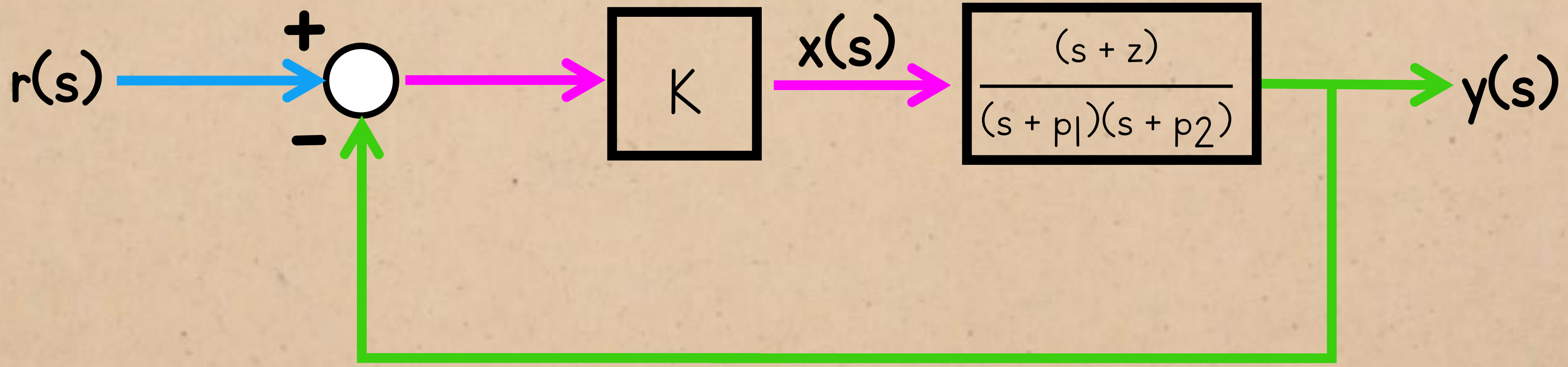
And that's what we get if we expand the original characteristic equation and increase the gain to infinity:

second-order system with one zero:

characteristic equation: $K(s + z) + (s + p_1)(s + p_2) = 0$



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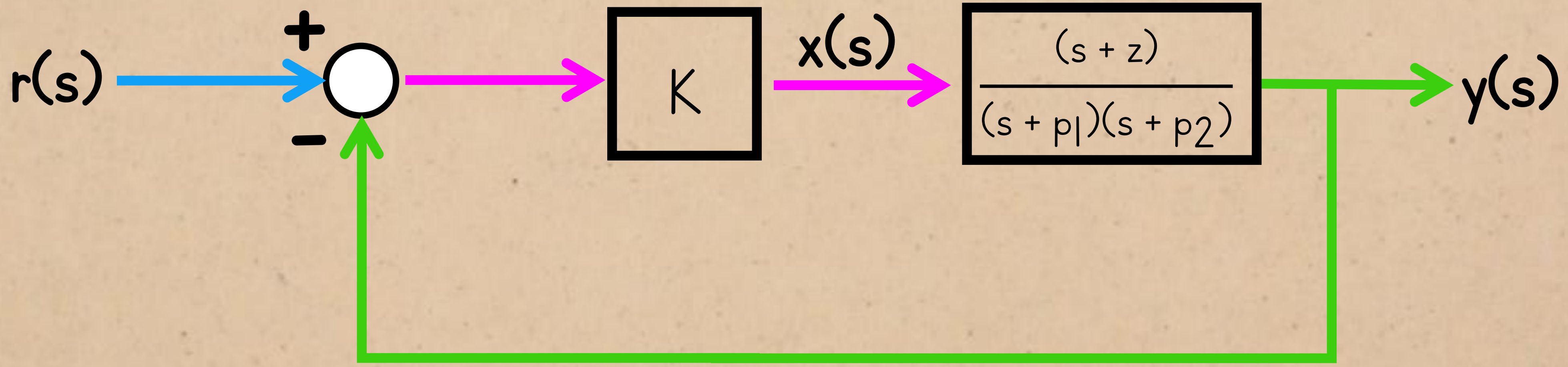
And that's what we get if we expand the original characteristic equation and increase the gain to infinity:

second-order system with one zero:

characteristic equation: $s^2 + (p_1 + p_2 + K)s + (p_1 p_2 + Kz) = 0$



And that's what we get if we expand the original characteristic equation and increase the gain to infinity:

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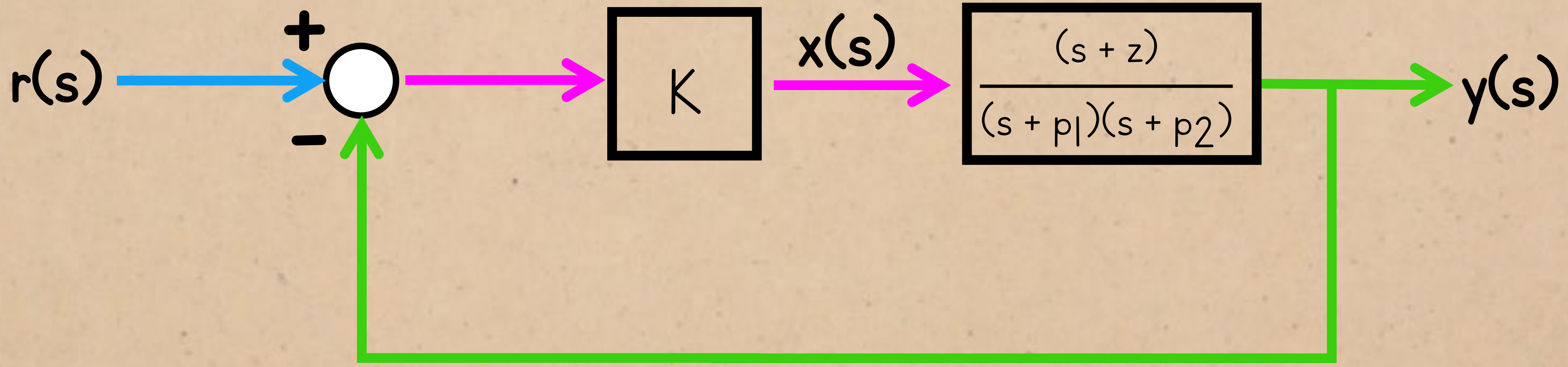
second-order system with one zero:

characteristic equation: $s^2 + (p_1 + p_2 + \infty)s + (p_1p_2 + \infty z) = 0$

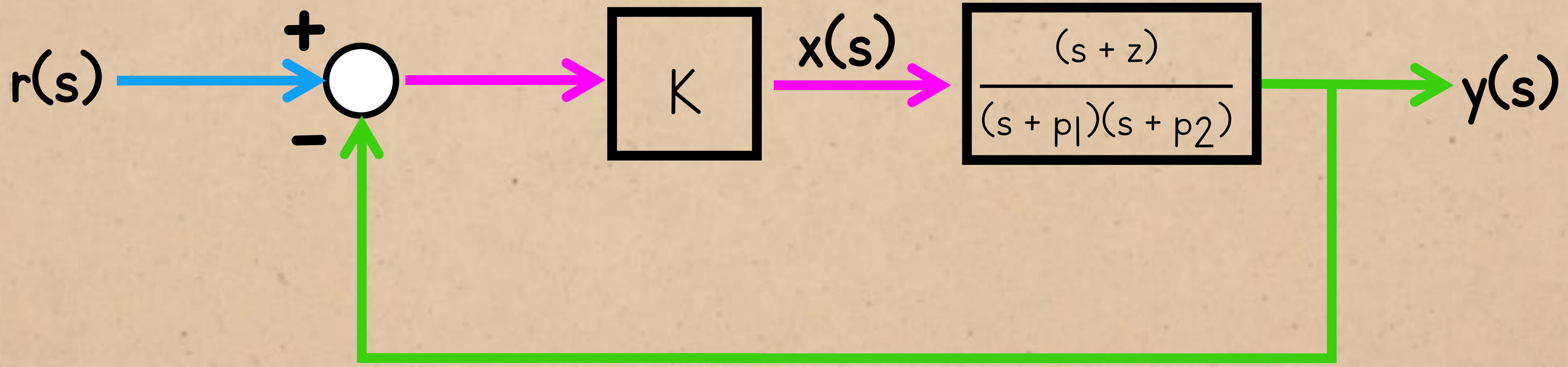


ROOT LOCUS

And that's what we get if we expand the original characteristic equation and increase the gain to infinity:

second-order system with one zero:

characteristic equation: $s^2 + \infty s + \infty z$



...But how do the poles get to their destinations?

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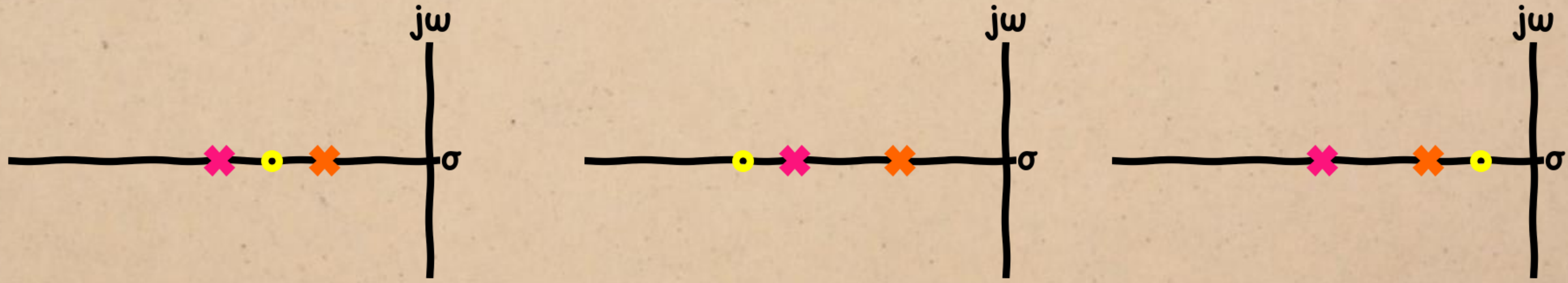
possible positions of one zero:



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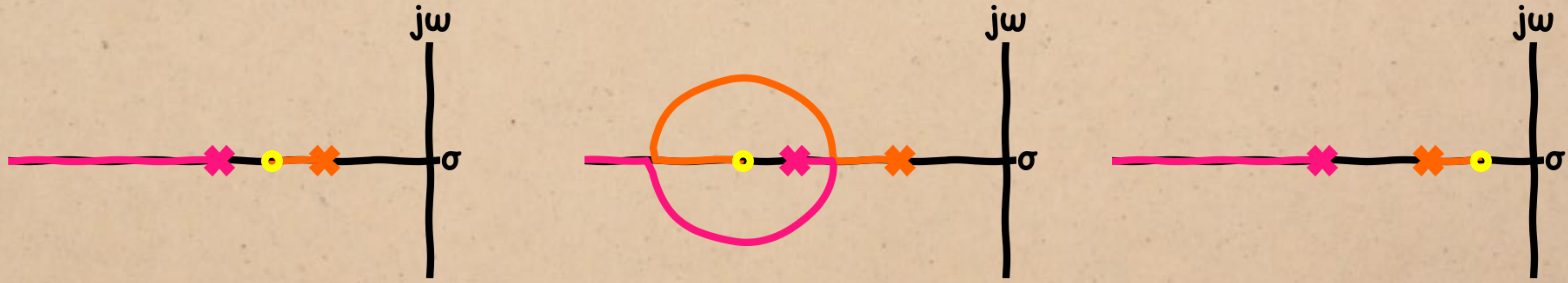
possible positions of one zero:



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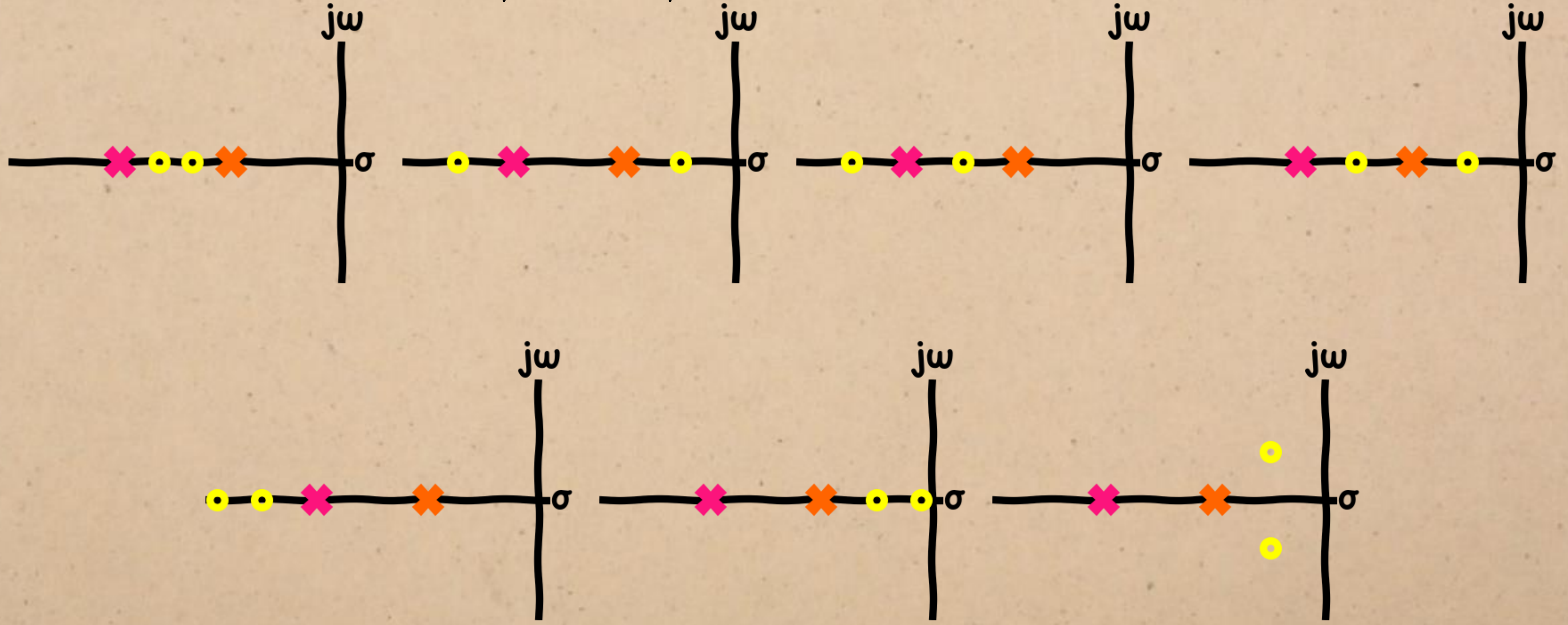
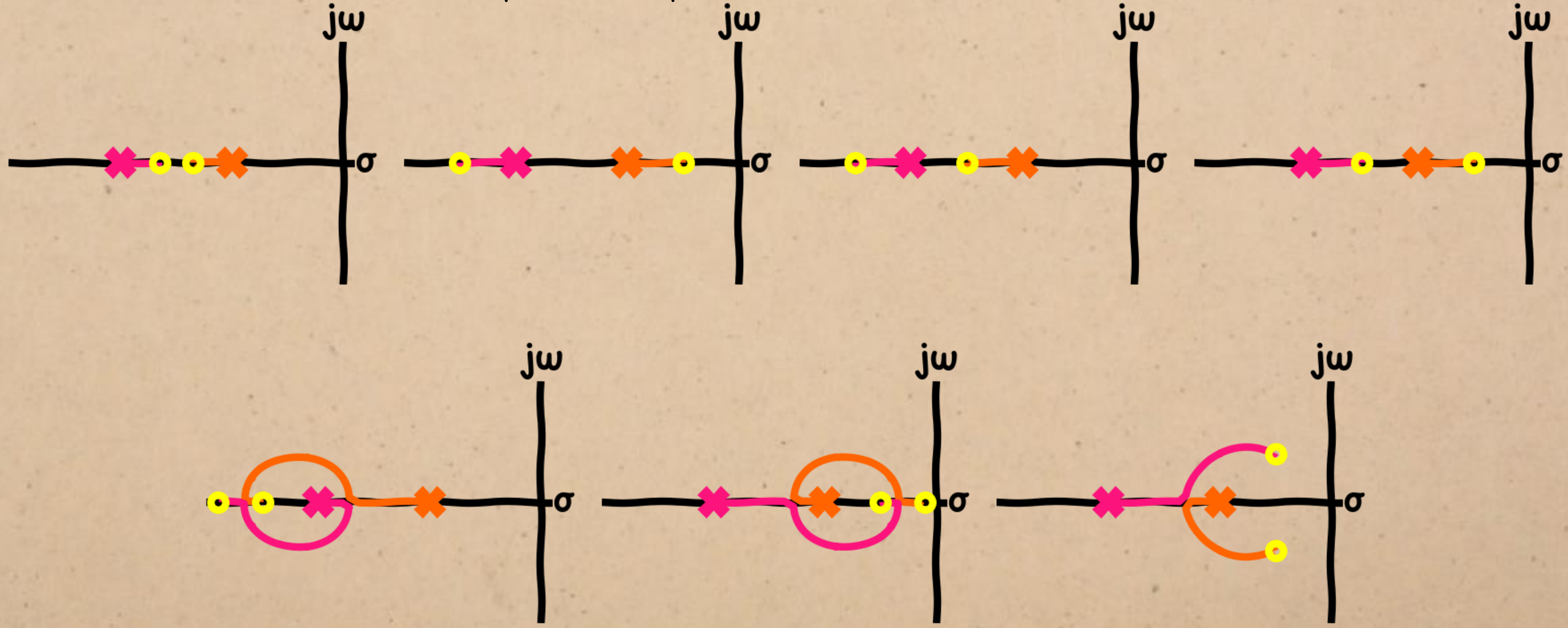
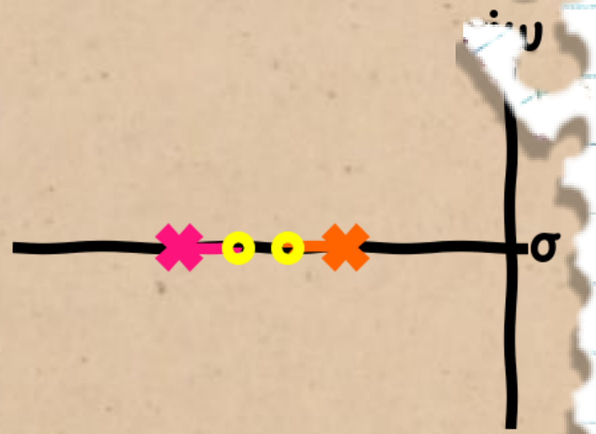
possible positions of two zeros:

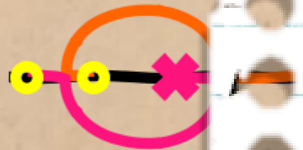


ROOT LOCUS

possible positions of two zeros:



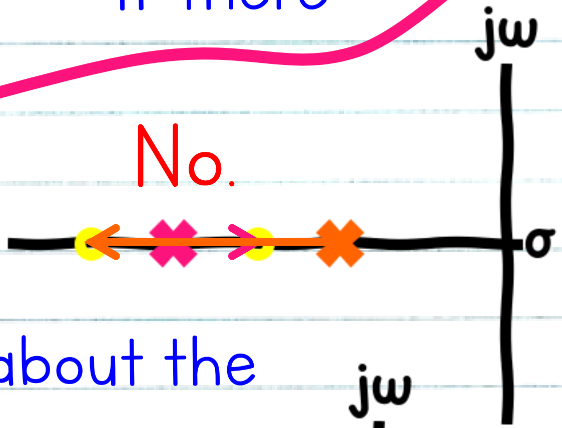




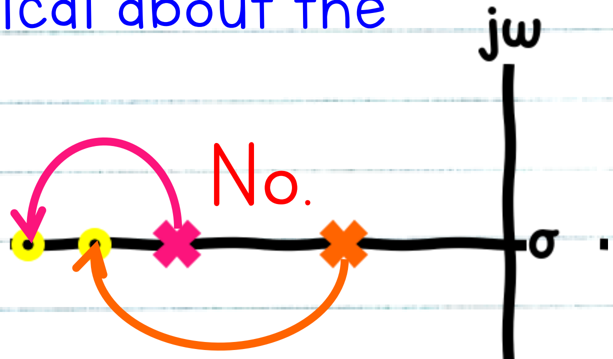
Observations

Closed-loop poles move from the open-loop poles to the open-loop zeros, or $-\infty$ if there is no zero available.

Pole's paths may not overlap.



Root loci must be symmetrical about the real axis. (duh)



not complex conjugates

We only need to follow these two rules to predict the rough shape of the root loci.

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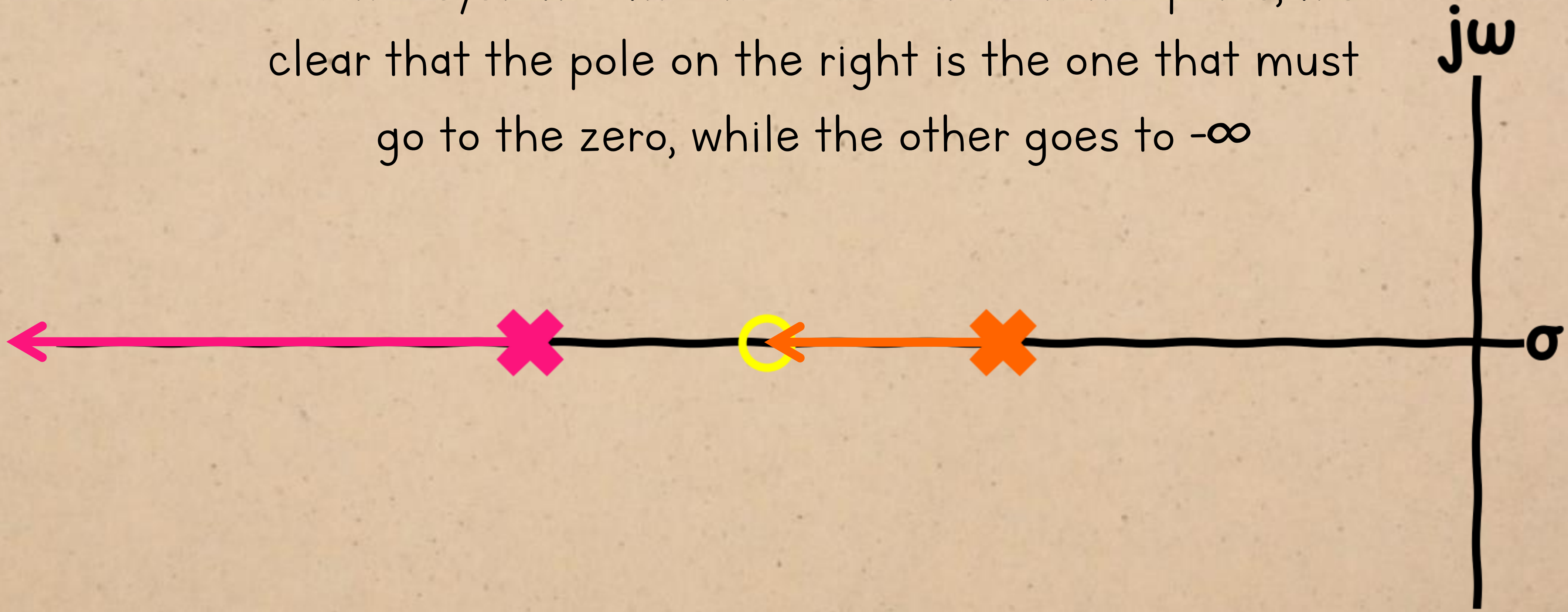
For the system with one zero between the poles, it's clear that the pole on the right is the one that must go to the zero, while the other goes to -∞



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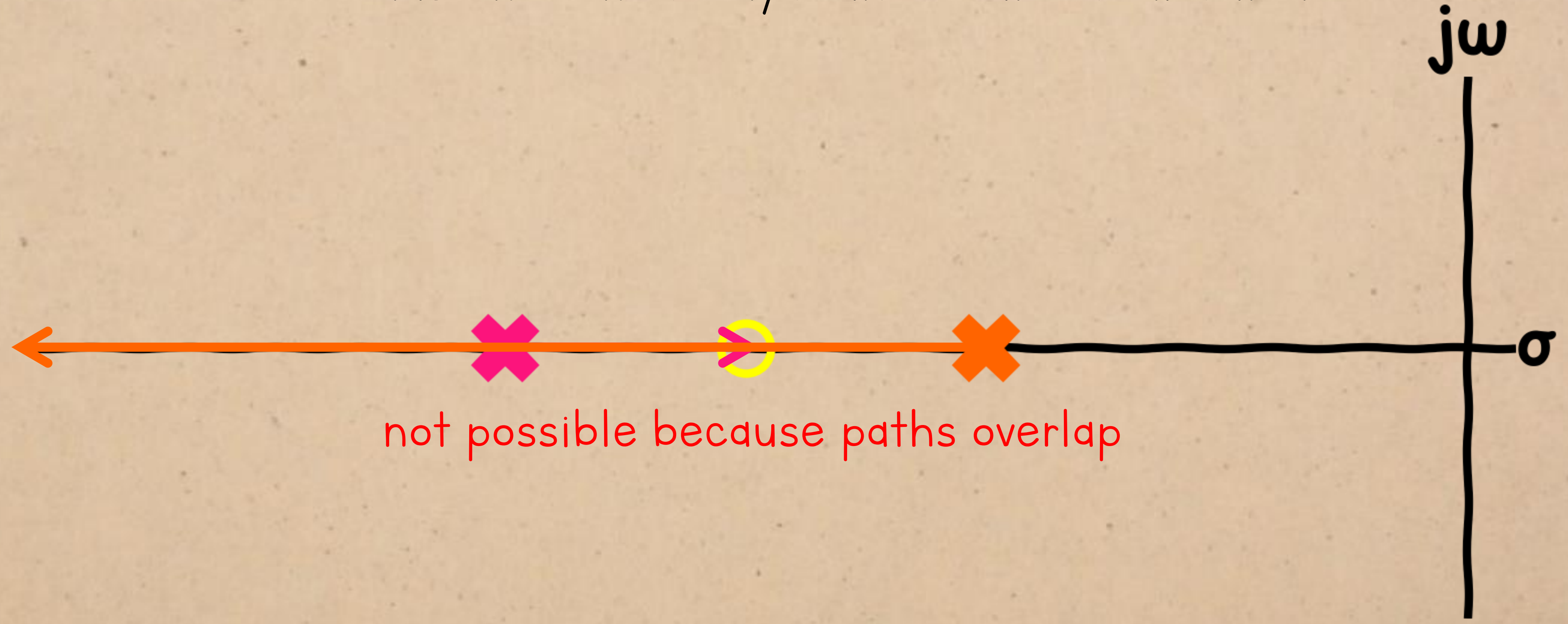
...because the other way round would break rule 1



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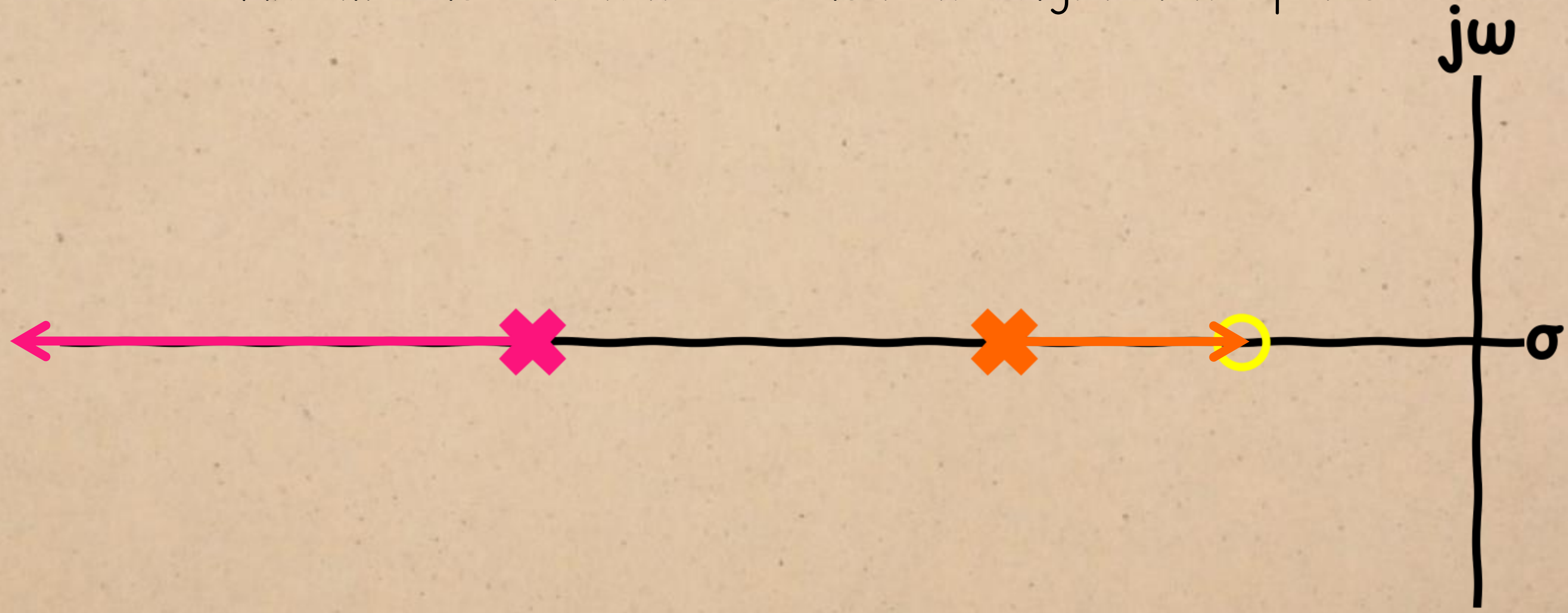
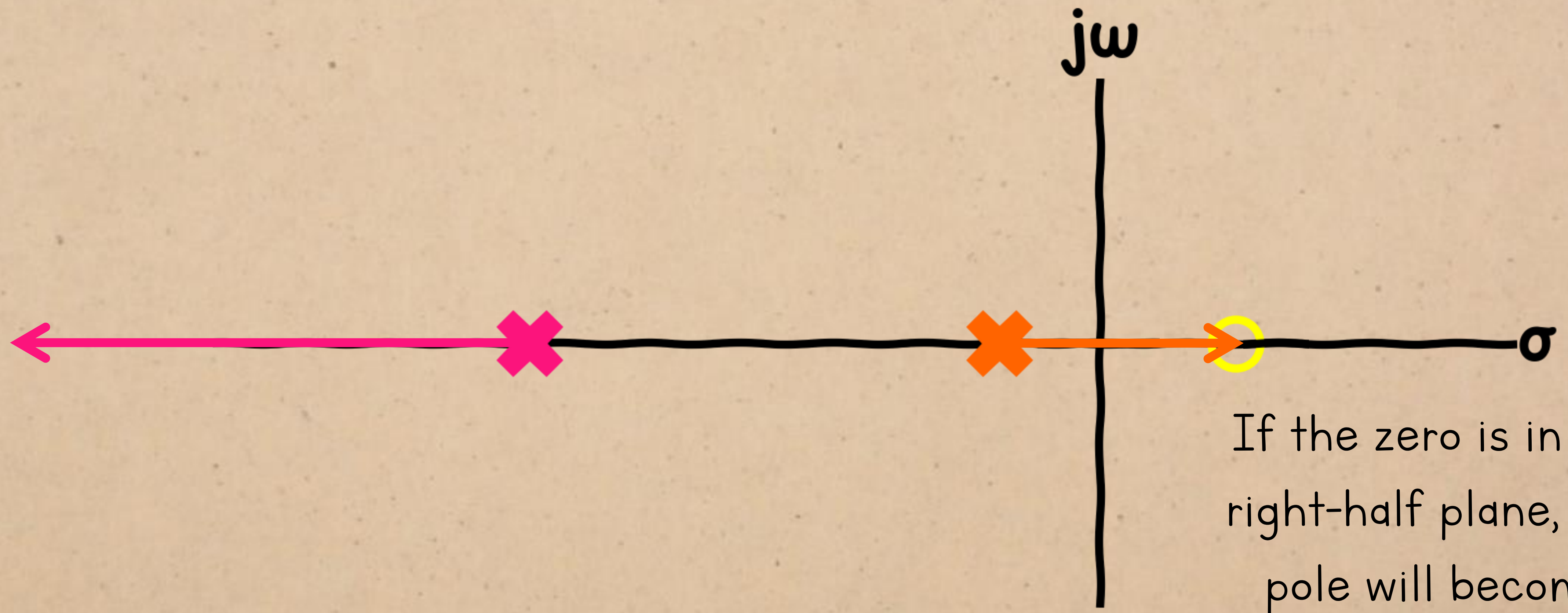
And likewise when the zero is to the right of the poles.



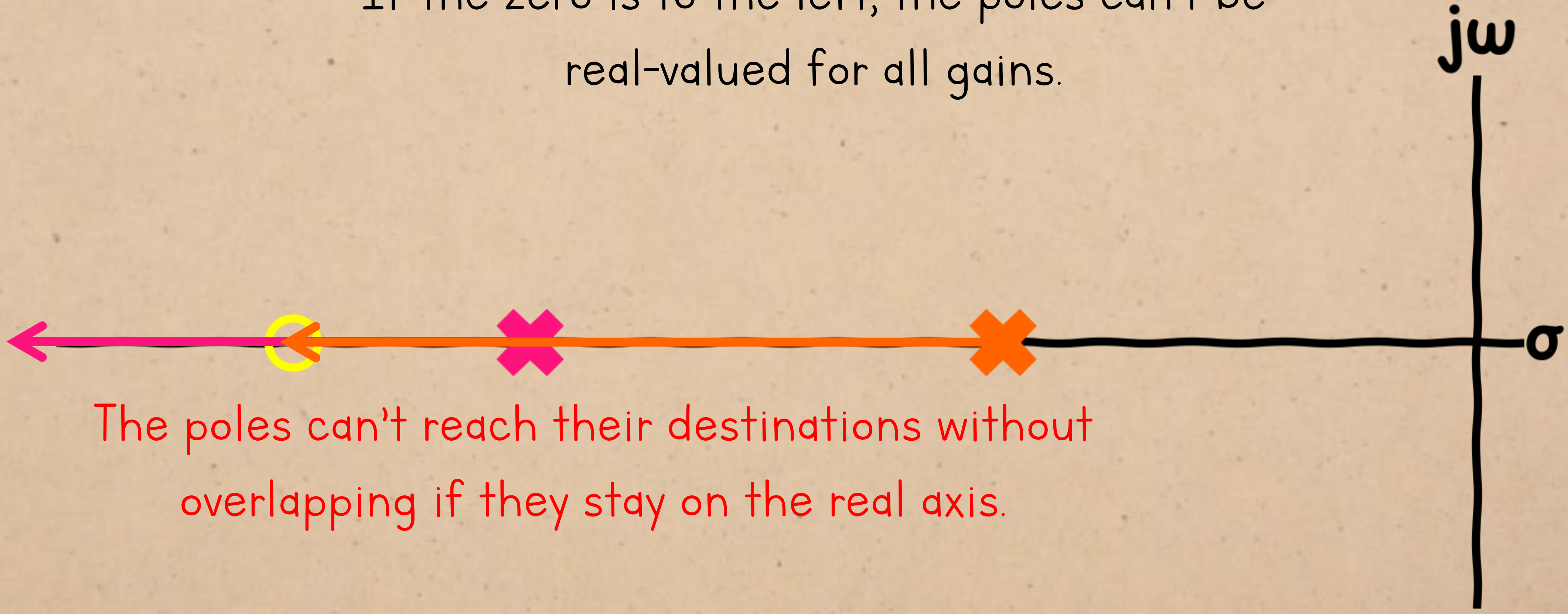
ROOT LOCUS



If the zero is in the right-half plane, this pole will become unstable for some gains

ROOT LOCUS

If the zero is to the left, the poles can't be real-valued for all gains.



The poles can't reach their destinations without overlapping if they stay on the real axis.

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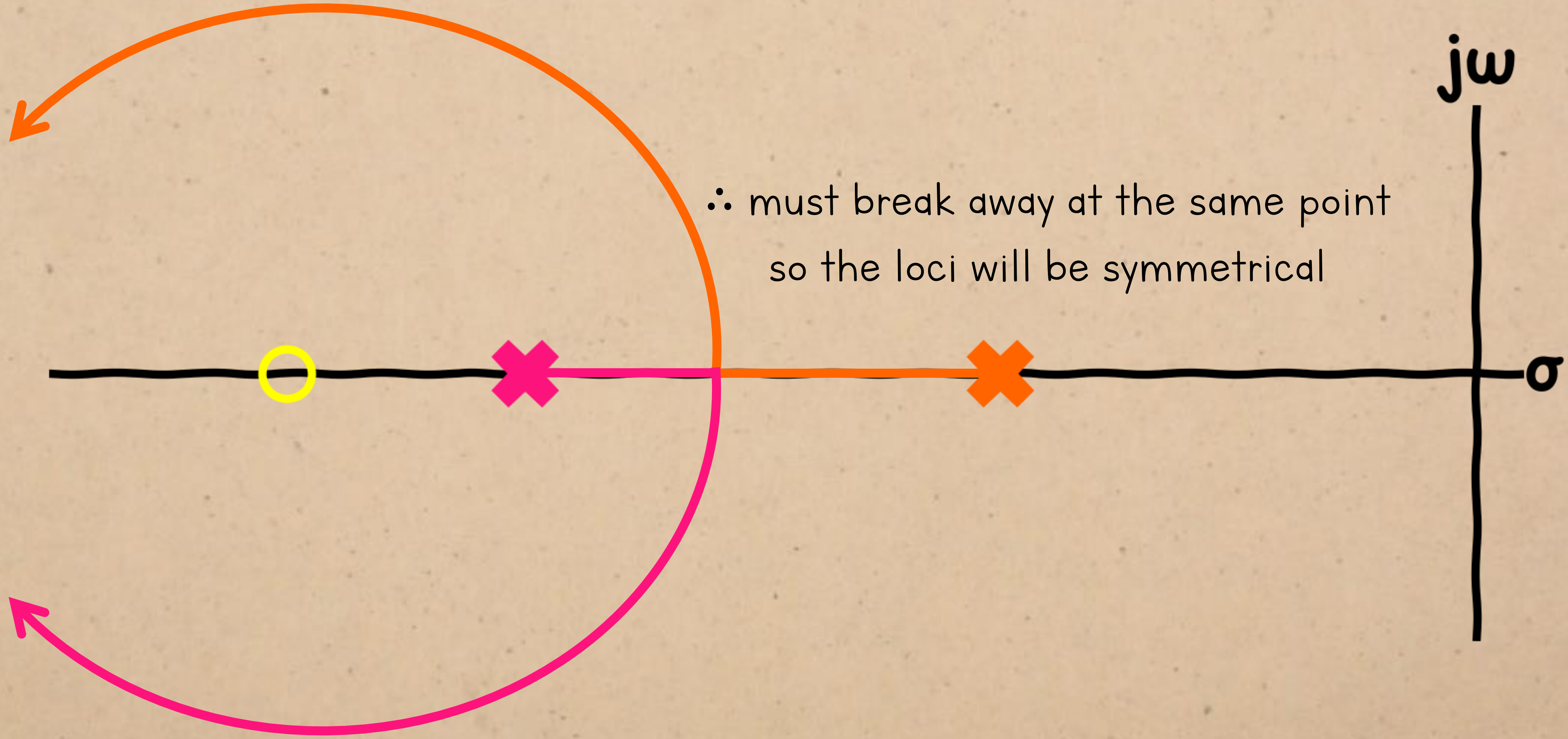
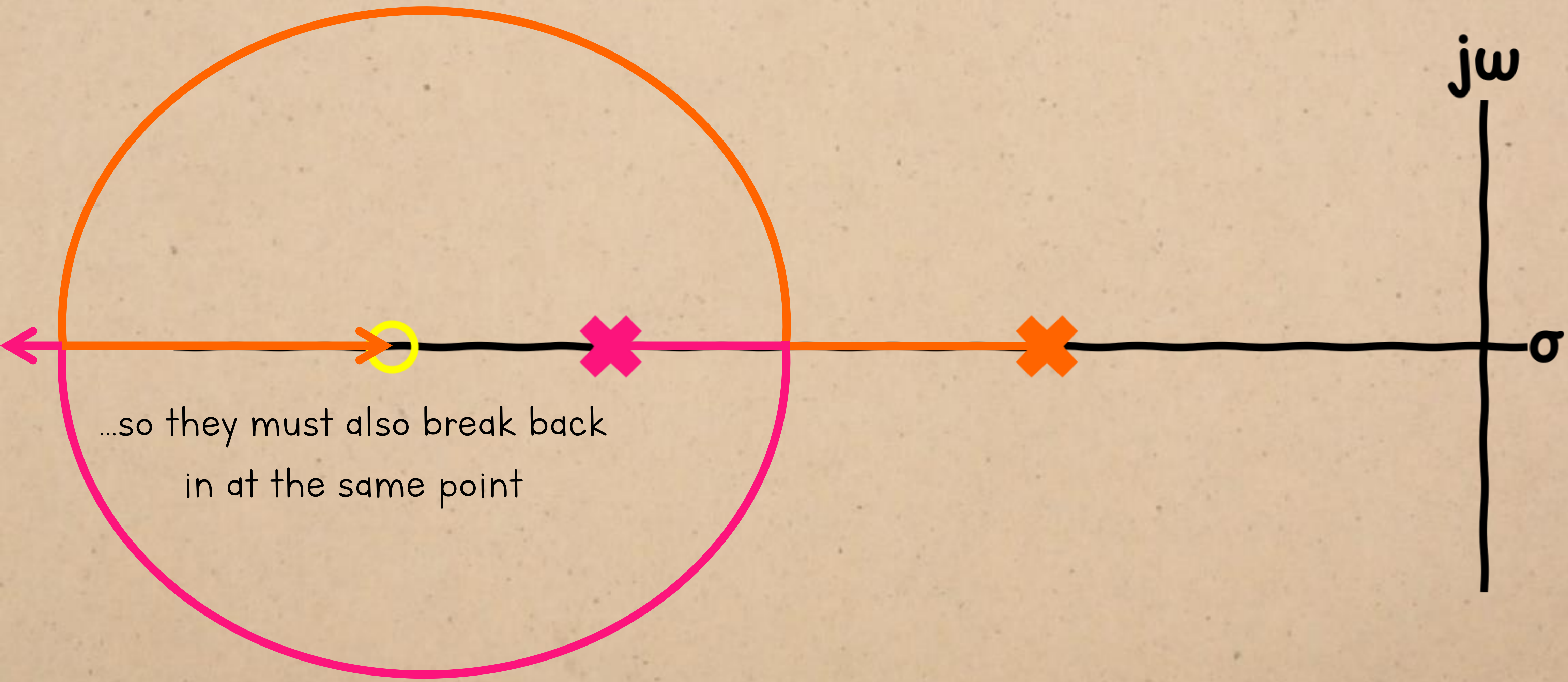


∴ must break away at the same point

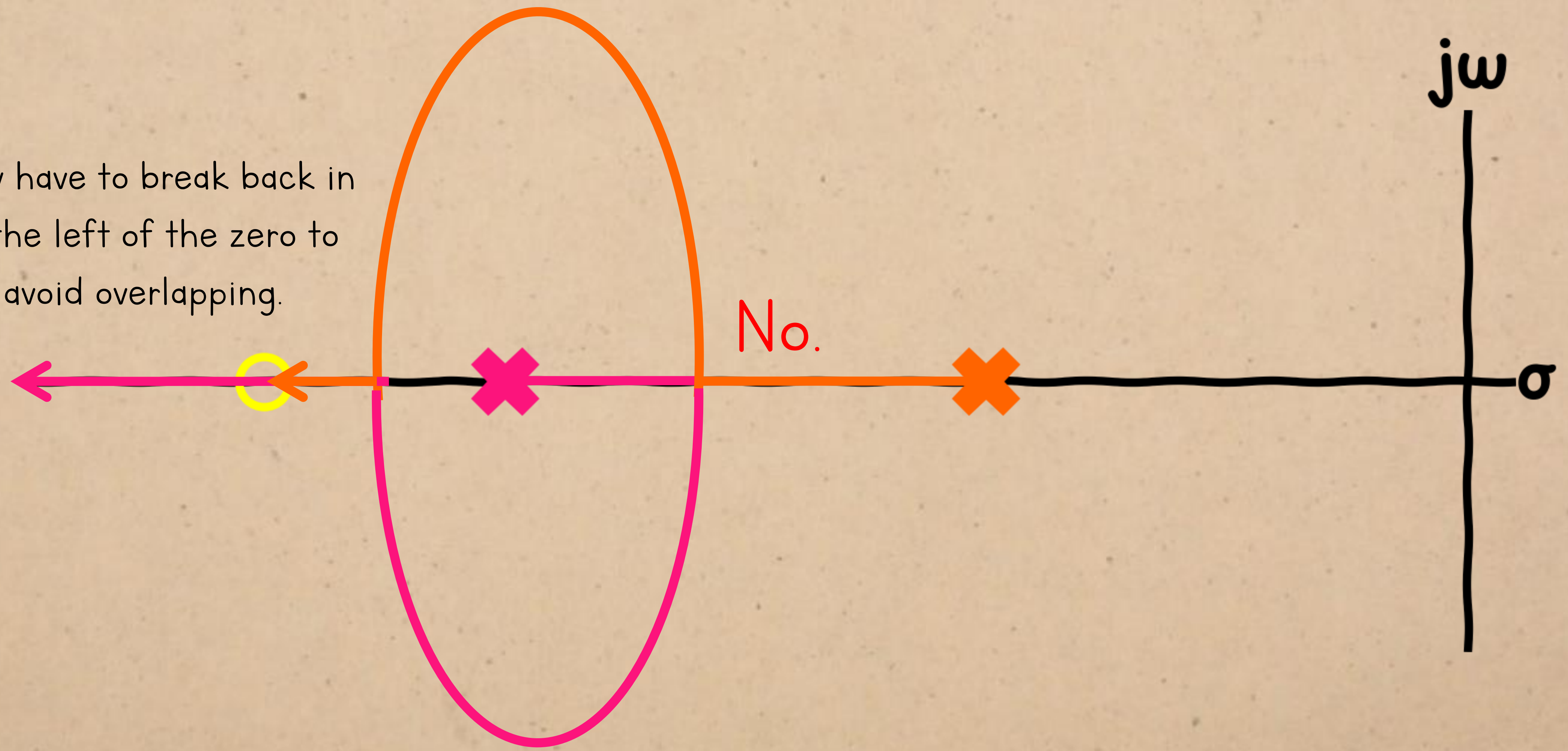
so the loci will be symmetrical

ROOT LOCUS



ROOT LOCUS

They have to break back in
to the left of the zero to
avoid overlapping.



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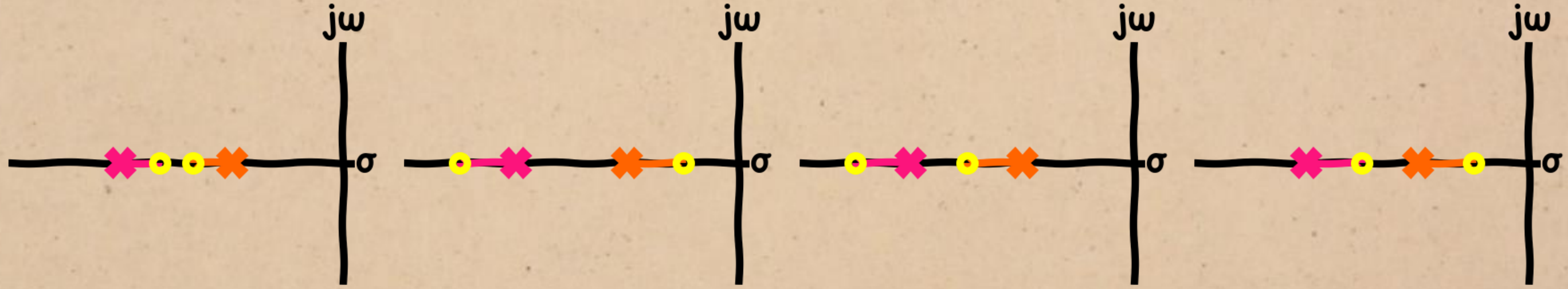
The paths are obvious for these four cases:



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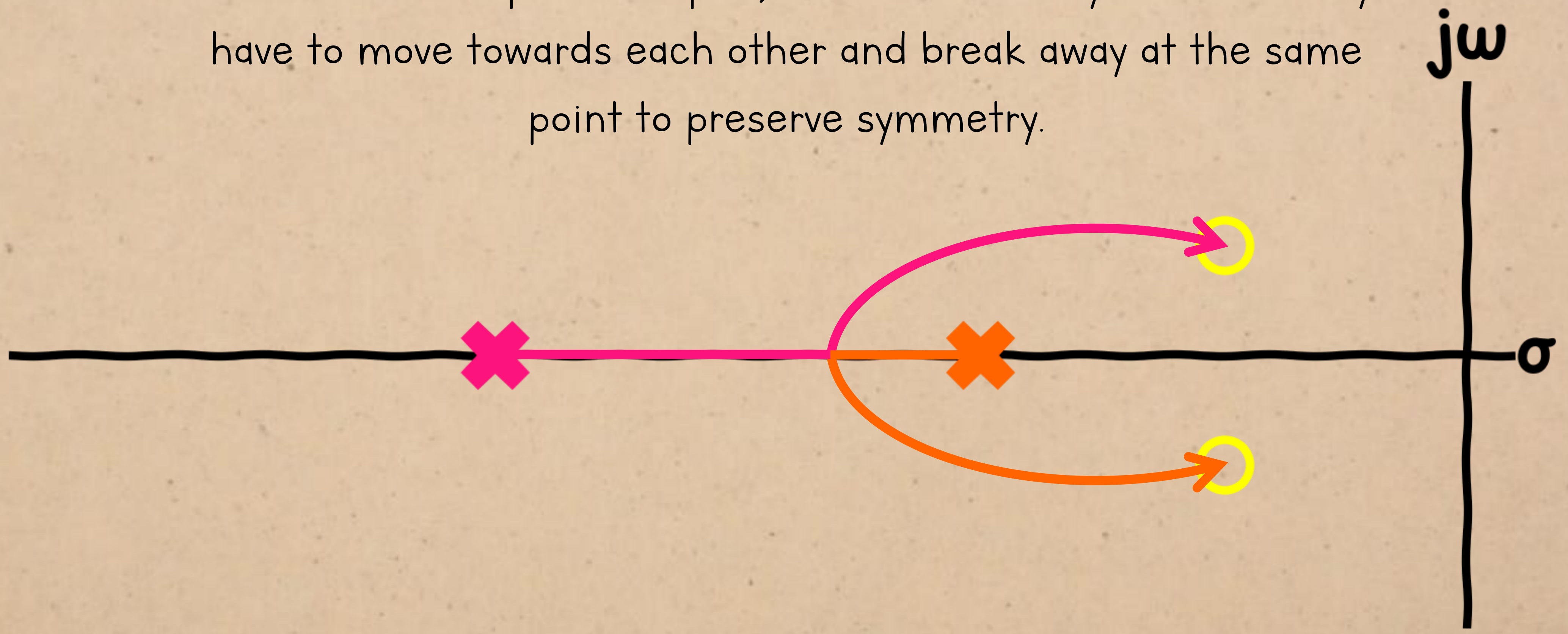
And for the complex zero pair, since we know by now that they have to move towards each other and break away at the same point to preserve symmetry.



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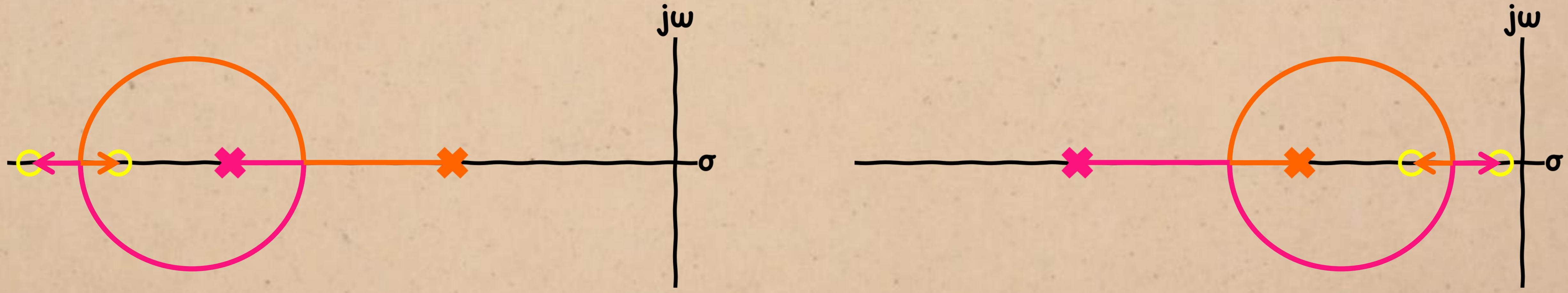
For these two cases, the poles have to become complex, and break in between the zeros so their paths won't overlap.



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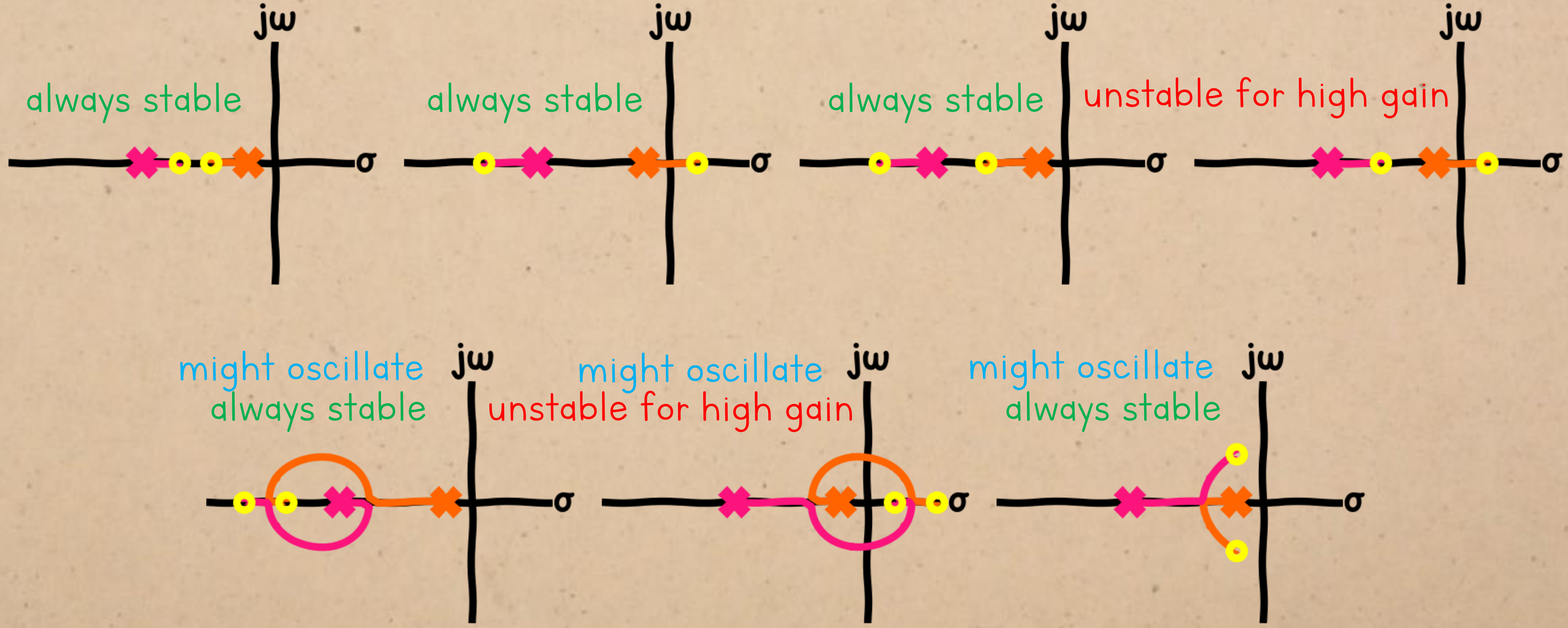
We now know enough to make some broad statements about what behaviour is possible for these systems in closed loop:



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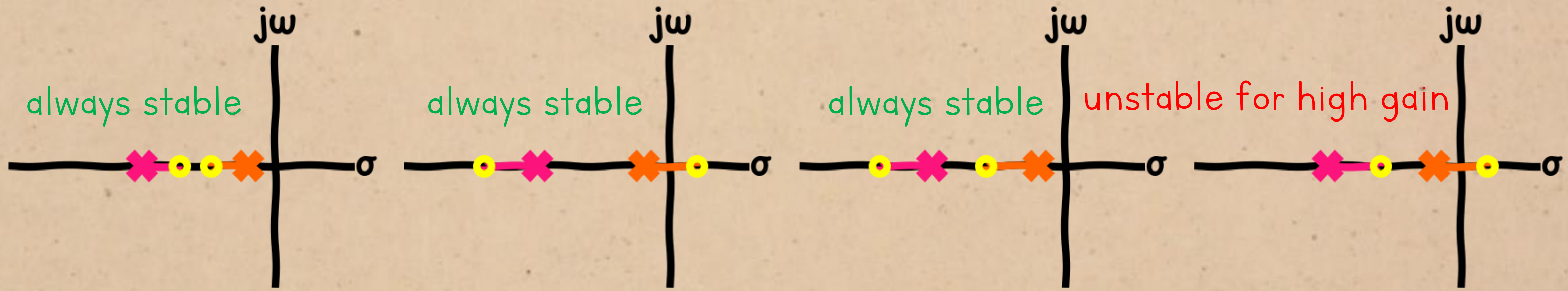
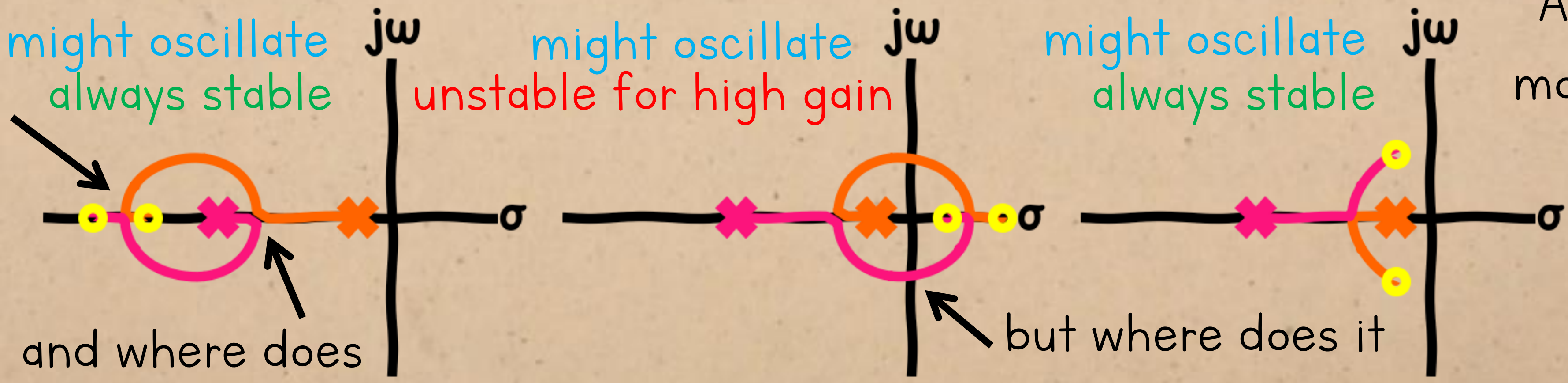
But we still need to find out:

Things we know about these systems:



And break back in?



and where does it break away?

but where does it cross the jω axis?

And what about more complicated systems?